FaceTrack

User Guide

A complete, step-by-step walkthrough of the Face Recognition Based Classroom Attendance System — the student, administrator and lecturer portals, every screen, every user flow, and a checklist for the day of the demonstration.

LIVE APPLICATION

https://attendace-system-david.vercel.app/

Document	System	Audience	Date
User Guide v2.0	**FaceTrack Attendance**	**Students, Administrators & Lecturers**	**October 2026**

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9	**Troubleshooting**	What to do when something does not work
10	**Demonstration checklist & quick reference**	What to do before and during a demonstration or defence

How to read this guide

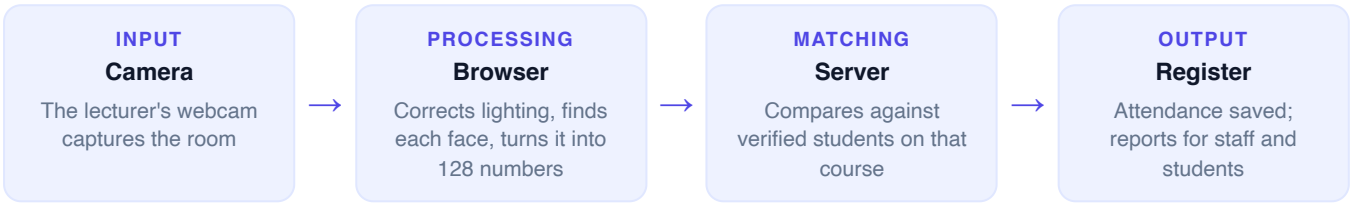
FaceTrack has **three portals**. Students register themselves online (section 4). The administrator checks each student's identity in person and captures their face (section 5). Lecturers take attendance with the camera (section 6). Section 7 joins them into end-to-end journeys. If you are about to demonstrate the system, go straight to the checklist in section 10.

1. What FaceTrack does

Attendance that is taken by the camera, not by a register passed around the hall.

In most lecture halls attendance is still taken on paper. It eats lecture time, the sheets get lost, and students sign for friends who are not in the room. FaceTrack replaces that with a camera at the front of the class: a student looks at it, is recognised from their own face, and is marked present in under a second.

How the parts fit together



Camera footage never leaves the computer

The browser converts each face into a list of 128 numbers — a "descriptor" — and only those numbers are sent to the server. No photograph of the classroom is ever uploaded, and a descriptor cannot be turned back into a picture of a face.

What it gives you

CAPABILITY	WHAT IT MEANS IN PRACTICE
Self-service registration	Students register online from anywhere and pick their own courses. Every field is chosen from a list, so records are consistent and the admin office types almost nothing.
Verified identities	A face is enrolled only in person, after the administrator checks the student against their JAMB result or admission letter. One face can never be enrolled under two matric numbers.
Fast, automatic capture	Enrolment takes three face samples by itself as soon as a face is in view — typically in under two seconds — and corrects poor lighting automatically.
No proxy attendance	A student must present their own face. Naming somebody is not enough — matching happens on the server, against templates the browser never sees.
No double entries	A student who stays in front of the camera for ten minutes still produces exactly one record.
Late tracking & locked registers	Anyone recognised after the cut-off is marked *late*; closing a session freezes the register.
Reports & eligibility	Attendance percentage per student against the course requirement (75% by default) — for staff as a register, and for each student on their own dashboard.

2. Before you start

Three things to check before your first lecture.

2.1 What you need

REQUIREMENT	DETAIL
Browser	Google Chrome or Microsoft Edge (recommended), or Safari, kept up to date.
Camera	Any built-in laptop webcam or USB camera. Needed only on the admin office PC (face enrolment) and the lecturer's PC (attendance). Students need no camera.
Internet	Needed to reach the site and save records. Face detection itself runs on the local machine.
Hardware acceleration	Leave it switched on in the browser (Chrome: *Settings → System → Use graphics acceleration when available*). Without it capture still works, but slower.

2.2 Preparing the camera engine — open the page early

The first time a camera screen is opened in a browser, about **12 MB** of face-recognition models download and the engine prepares itself. The camera button reads `Preparing…` and the panel shows what is happening (*Loading face detector…*, *Preparing the camera engine (first time only)…*). This usually takes 5–30 seconds and happens **once per browser session**: every capture after it is fast.

> **Open the camera page a minute before you need it**
>
> Administrator: open a student's **Verify & enrol** page while the student is still walking to your desk. Lecturer: open the **live capture** screen while students settle. When the button changes to **Start face capture** / **Start capture**, the engine is ready.

Under the camera, a line reports the engine's mode. *Face engine: GPU-accelerated* is the normal, fast mode. *CPU mode — capture will be slower* means the browser's hardware acceleration is off; capture still works (about 7 seconds for an enrolment instead of under 2), and switching acceleration back on restores full speed.

2.3 The camera only works on a secure address

ADDRESS	CAMERA
`https://attendace-system-david.vercel.app/`	✓ Works
`http://localhost:3000` (running it yourself)	✓ Works
`http://192.168.x.x:3000` (a LAN address)	✗ Blocked by the browser

3. Portals, accounts & passwords

Three kinds of user, each with their own portal and their own job.

3.1 The three portals

Student portal	Administrator portal	Lecturer portal
Registers online from anywhere: personal and academic details, passport photograph, password. Registers their courses and follows their own attendance and exam eligibility. **Cannot** enrol a face.	The department officer. Verifies each student in person against their JAMB result or admission letter and captures their face. Corrects records, creates courses, assigns lecturers, manages staff, monitors lectures.	Sees only the courses assigned to them. Starts a lecture session, takes attendance with the camera, marks stragglers manually, closes the register, reads the reports.

Why can students not capture their own face?

Impersonation. If a face could be enrolled from home, a student could enrol a friend's face under their own matric number and send the friend to every lecture. So registration is self-service, but the face is captured **only in person by the administrator, after an identity check** — and the system refuses any face that is already enrolled under another matric number. Every rule is enforced on the server, not just by hiding buttons.

3.2 The sign-in screen EVERYONE `/login`

1. Open `https://attendace-system-david.vercel.app/`. You are taken to the sign-in screen.
2. Choose your portal with the tabs: **Student** (the default), **Lecturer** or **Administrator**.
3. Students enter their **matric number**; staff enter their **email address**. Then the password.
4. Press **Sign in**. Each portal opens its own dashboard. New students use **Create your account** under the form.

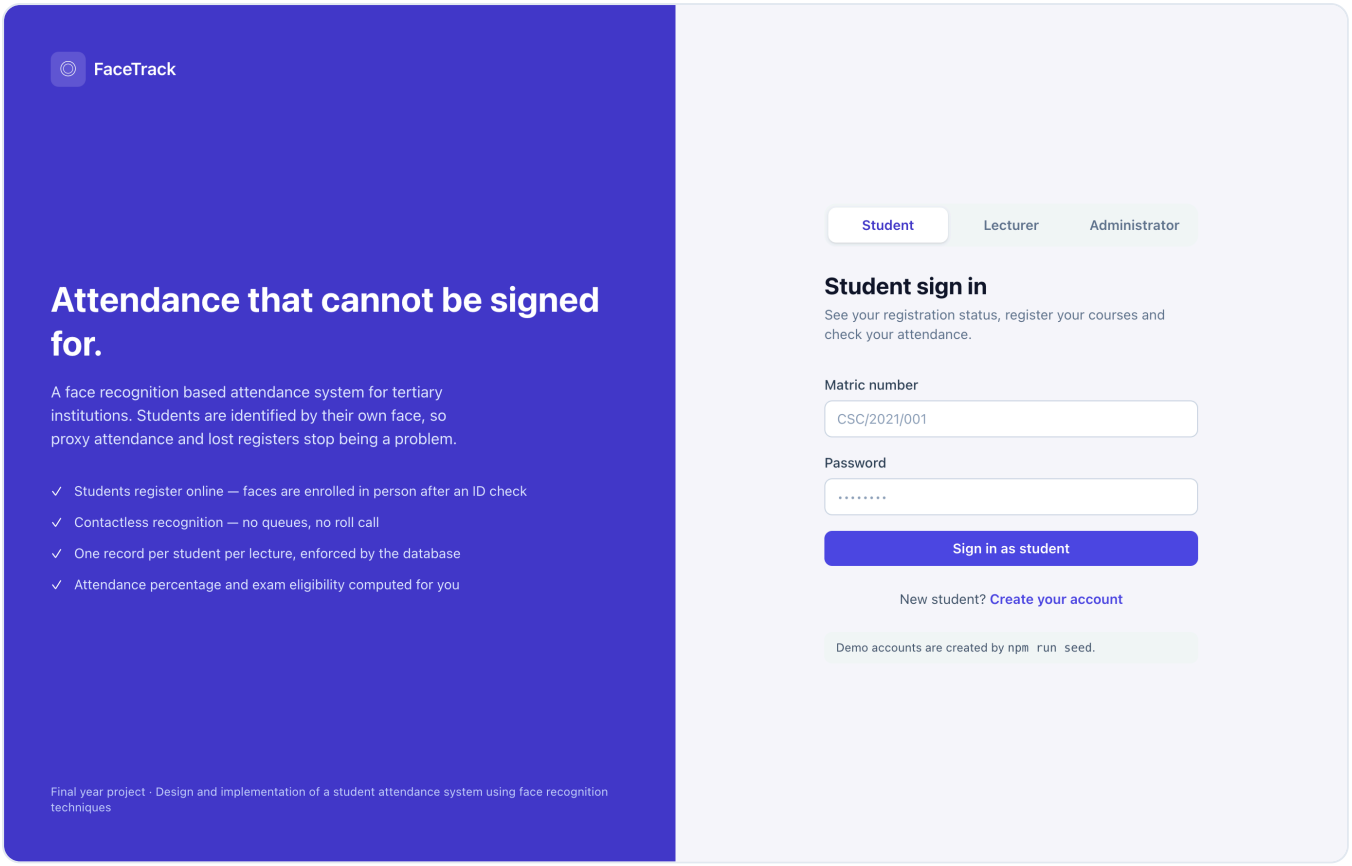


Figure 1. The sign-in screen on the **Student** tab, with the link for new students to create an account.

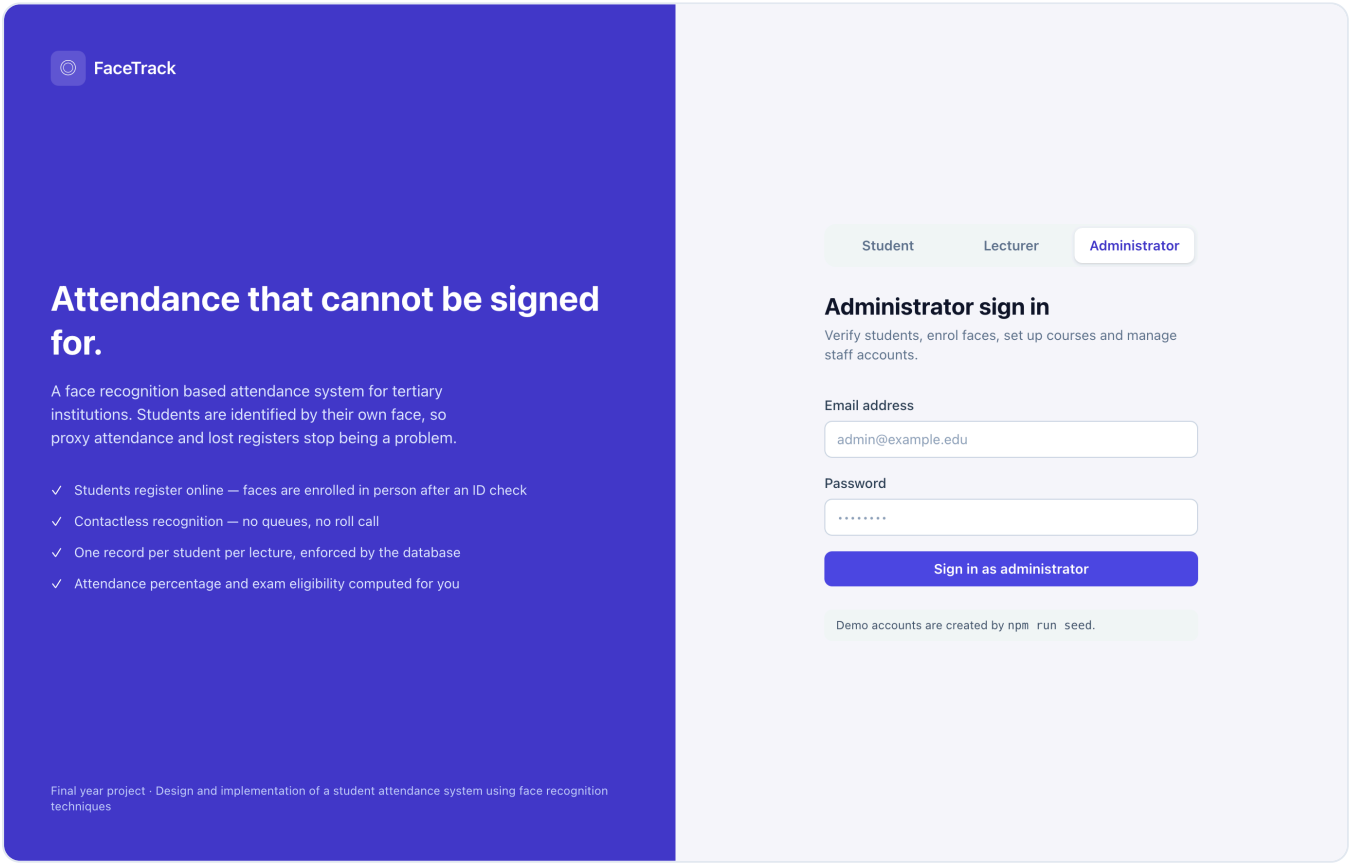


Figure 2. The **Administrator** tab. An account only works on its own tab; using the wrong one tells you which tab to use.

3.3 Demonstration accounts

Student tab

Matric no.
`CSC/2021/001`

Password
`student1234`

Any seeded student, `CSC/2021/001` to `CSC/2021/010`, uses the same password.

Administrator tab

Email
`admin@example.edu`

Password
`admin1234`

Signs in as Dr. Grace Adeyemi.

Lecturer tab

Email
`lecturer@example.edu`

Password
`lecturer1234`

Signs in as Mr. Emeka Nwosu, lecturer of the CSC courses.

These are demonstration passwords

They are deliberately simple so the system can be shown and marked. Before real use, create fresh staff accounts with strong passwords and remove these.

3.4 What each role can do

ACTION	STUDENT	ADMINISTRATOR	LECTURER
Create an account online	✓	✗	✗
Register / drop own courses	✓	—	✗
See own attendance & eligibility	✓	—	—
Verify identity & capture a face	✗	✓	✗
Correct a student's details, reset a password	✗	✓	✗
Create courses, assign lecturers, manage class lists	✗	✓	✗
Create staff accounts	✗	✓	✗
Start a session, capture, mark manually, close	✗	✗	✓ **own courses**
View registers and course reports	✗	✓ **all**	✓ **own courses**

Signing out

Your name is at the bottom of the sidebar with a Sign out button. The coloured label at the top — *Student*, *Administrator* or *Lecturer portal* — always shows where you are. Sessions expire after 8 hours.

4. The student portal

Registering from anywhere, and following your own attendance.

Sidebar: Dashboard · Course registration · My attendance

Sign in on the **Student** tab with your matric number.

4.1 Registering online STUDENT `/register`

Open the site, choose the **Student** tab and click **Create your account**. The form has three parts, and almost every answer is picked from a list.

1. **Personal details** — full name (surname first), gender (list), phone number and email address.
2. **Academic details** — choose your **faculty**; the **department** list then shows only that faculty's departments. Choose your **level** and **year of entry**.
3. **Matric number** — the first parts are filled in for you from your department code and year of entry (for example `CSC/2022/`). Type only the final digits; the full number is shown underneath.
4. Type your **JAMB registration number** exactly as printed on your result slip.
5. **Upload a passport photograph** — a clear, recent photo of your face. The administrator compares it with you in person.
6. Choose a **password**, confirm it, and press **Create my account**. You are signed in straight away.

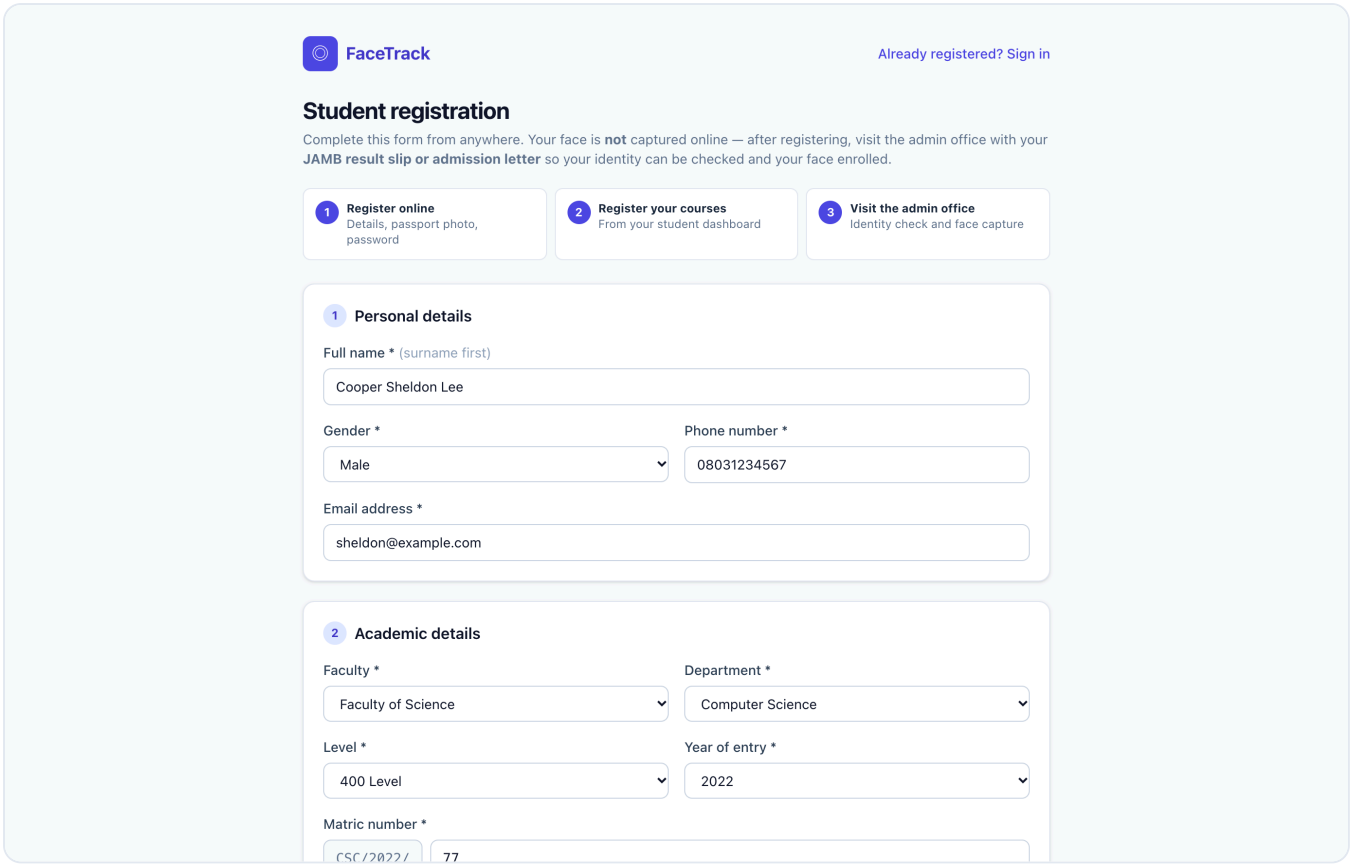


Figure 3. Online registration. Gender, faculty, department, level and year of entry are all dropdowns.

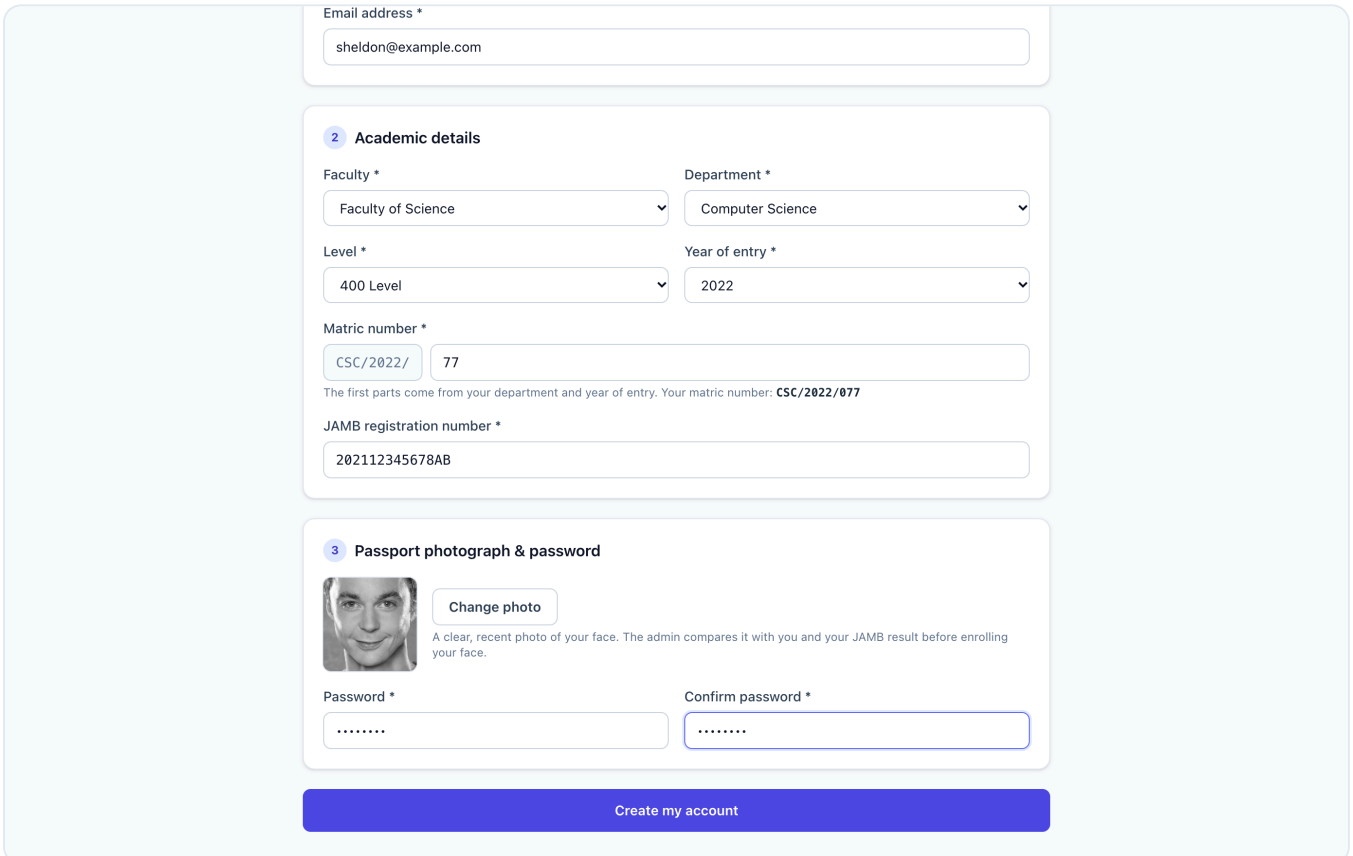


Figure 4. The matric number is assembled from the department and year (`CSC/2022/` + `77` → **CSC/2022/077**); then the JAMB number, passport photograph and password.

> **Your face is not captured online**
>
> That happens only at the admin office (section 4.4). Registration is refused if the matric number or JAMB number is already registered.

4.2 Student dashboard STUDENT `/dashboard`

A three-step progress tracker shows what is done and what is next, beside your record and passport photo. Until your face is enrolled, an amber banner reminds you to visit the admin office.

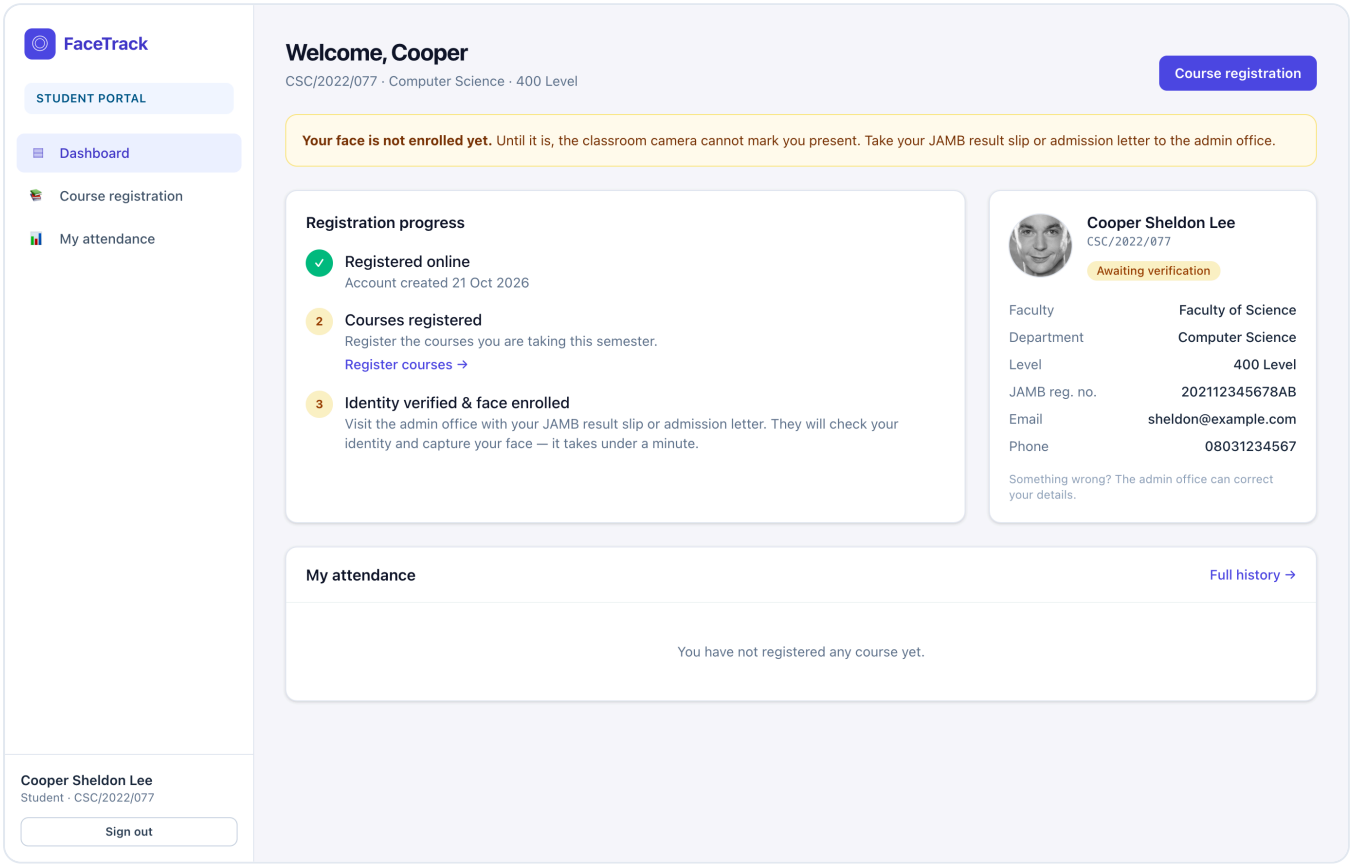


Figure 5. Straight after registering: step 1 done, steps 2 and 3 still to do.

4.3 Course registration STUDENT `/my-courses`

1. Open **Course registration**. *Courses on offer* is already filtered to your department and level; change the **Department**, **Level** or **Semester** dropdowns to see others.
2. Press **Register** beside each course you are taking. It moves to *My registered courses*, with your total units.
3. **Drop** removes a course — allowed only until attendance has been recorded for you on it.

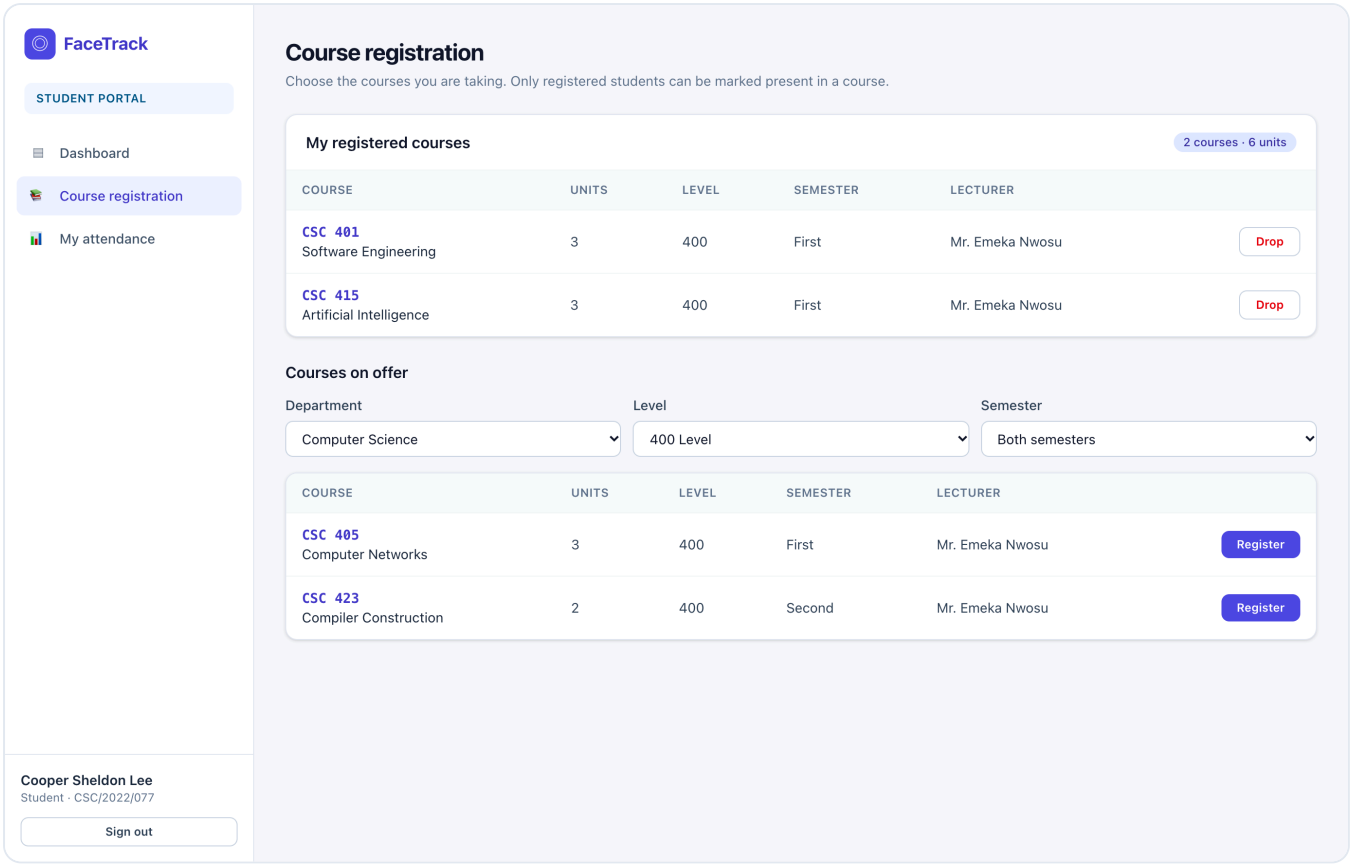


Figure 6. Two courses registered (6 units). Only students registered on a course can be recognised in its lectures.

4.4 Visiting the admin office

Take your **JAMB result slip** or **admission letter** (or school ID card) to the admin office. The administrator compares you with the photograph on the document and with your uploaded passport, then captures your face with the camera — it takes under a minute. After that your dashboard shows **Verified** and the classroom camera can mark you present.

4.5 Attendance & exam eligibility STUDENT

`/dashboard · /my-attendance`

Once verified, the dashboard lists each registered course with classes held, classes attended (and how many late), your percentage, and whether you meet the requirement to sit the exam. **My attendance** lists every lecture you were marked in, filterable by course.

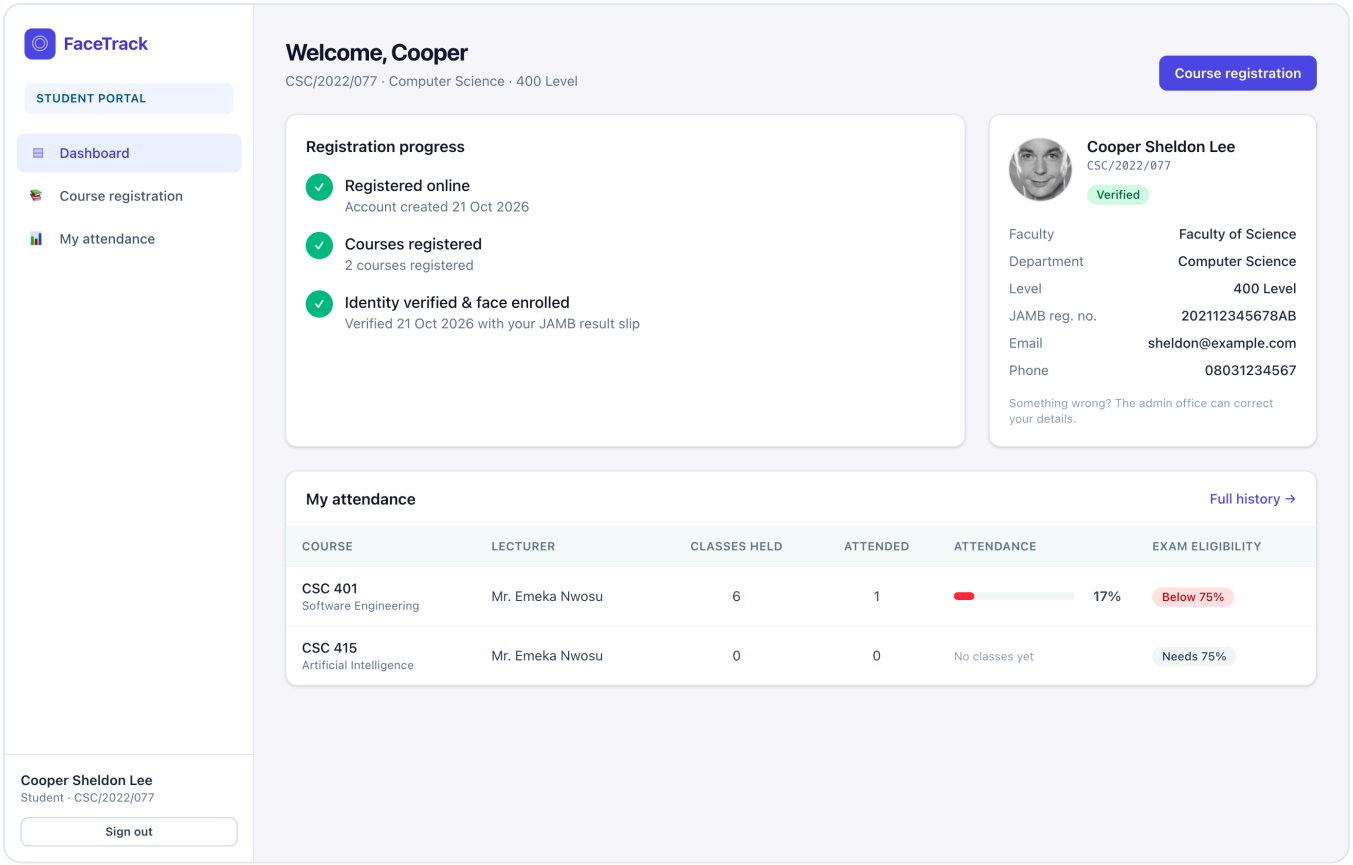


Figure 7. All three steps complete. CSC 401 shows 17% because five lectures took place before this student registered the course.

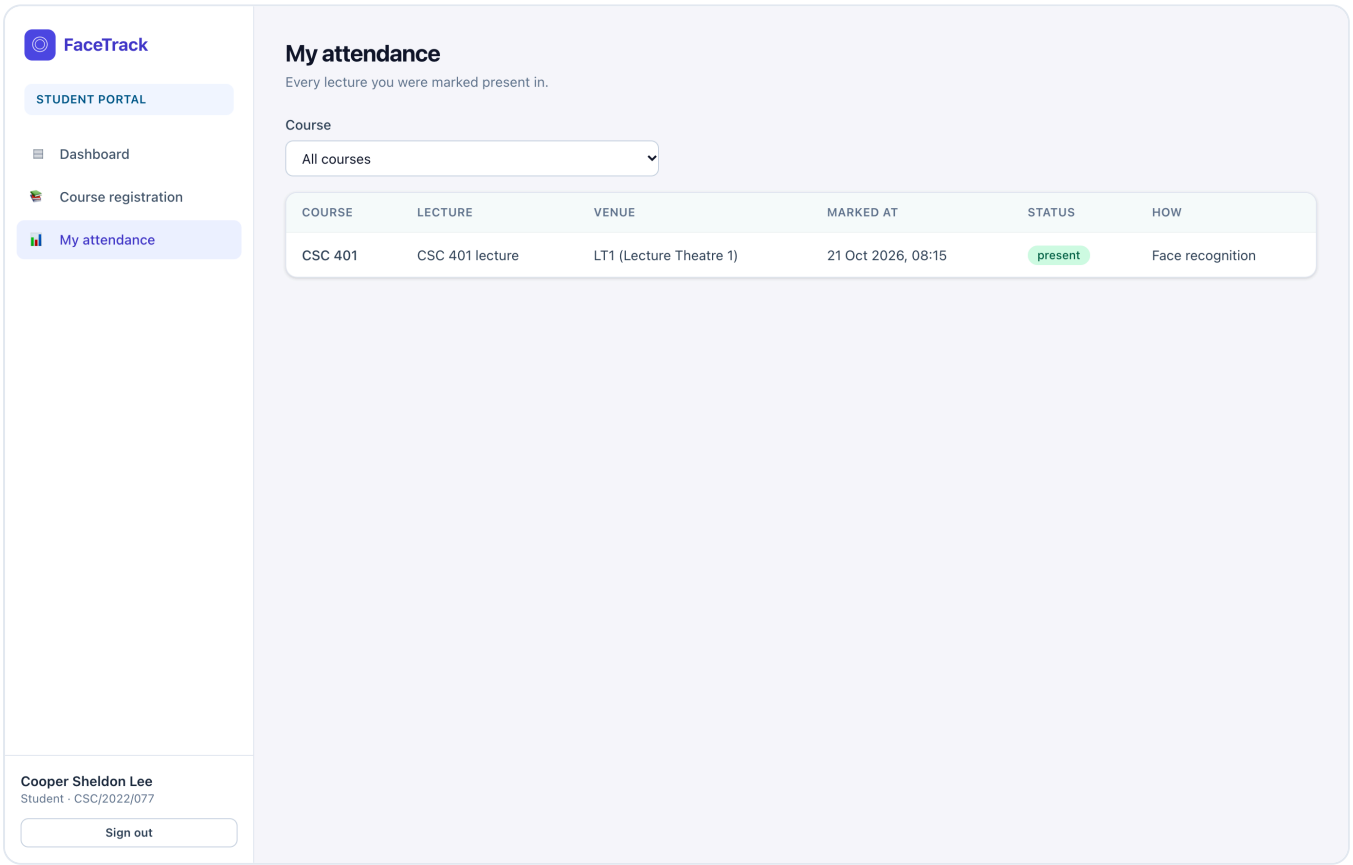


Figure 8. My attendance — each record shows whether it came from face recognition or was marked by the lecturer.

5. The administrator portal

Verifying students, capturing faces, and keeping records correct.

Sidebar: Dashboard · Students · Courses · Staff accounts · Session monitor · Attendance reports

Sign in on the **Administrator** tab.

5.1 Administrator dashboard ADMIN `/dashboard`

A department-wide overview: how many students are **awaiting verification**, courses without a lecturer, lecturer accounts, attendance marked today and classes in progress. **Verify students** (top right) shows how many are waiting.

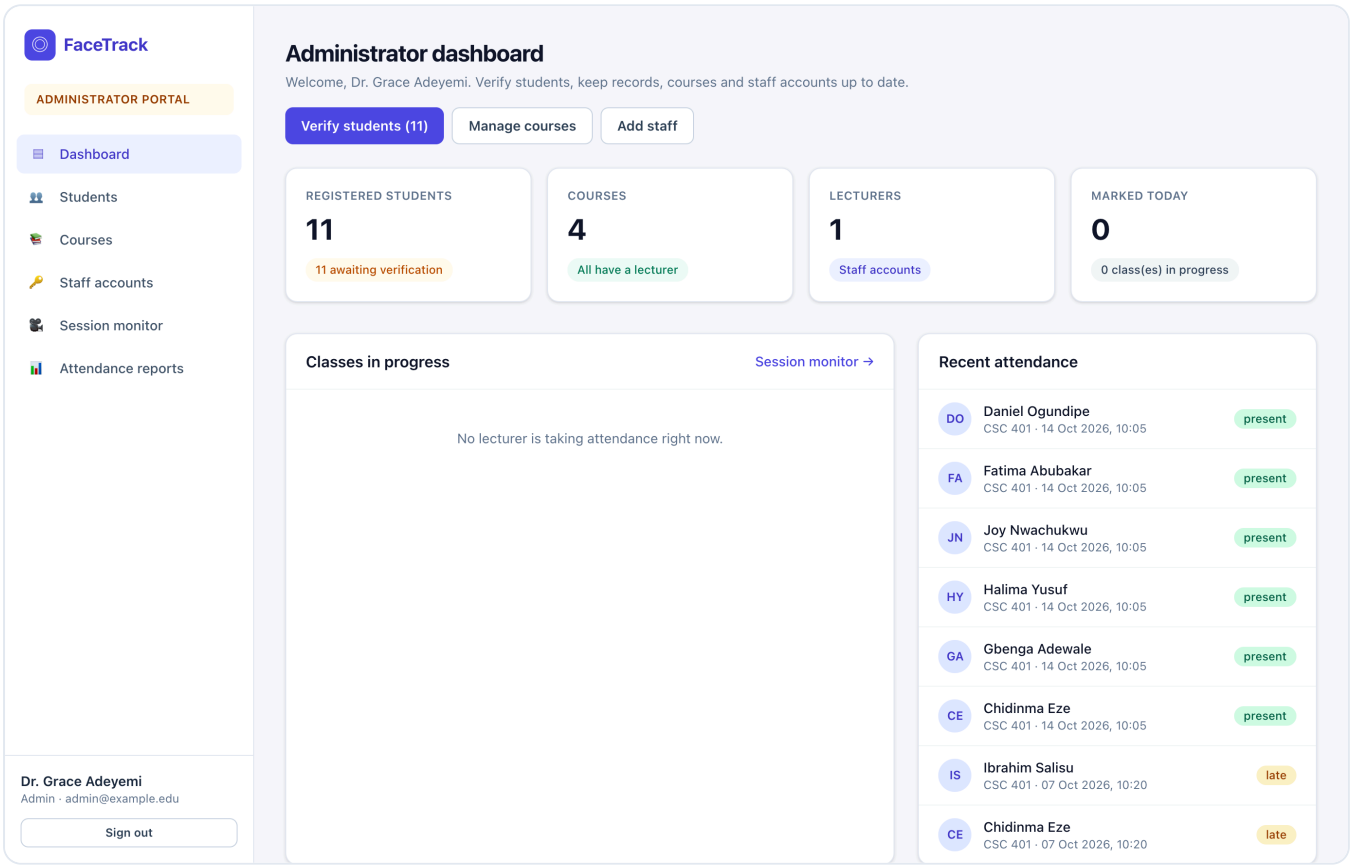


Figure 9. The administrator dashboard.

5.2 Students & the verification queue ADMIN `/students`

Students appear here as soon as they register online. The page opens on **Awaiting verification**; the other tabs show **Face enrolled** and **All students**. Filter with the search box and the **Department** and **Level** dropdowns. Press **Verify & enrol** when a student arrives.

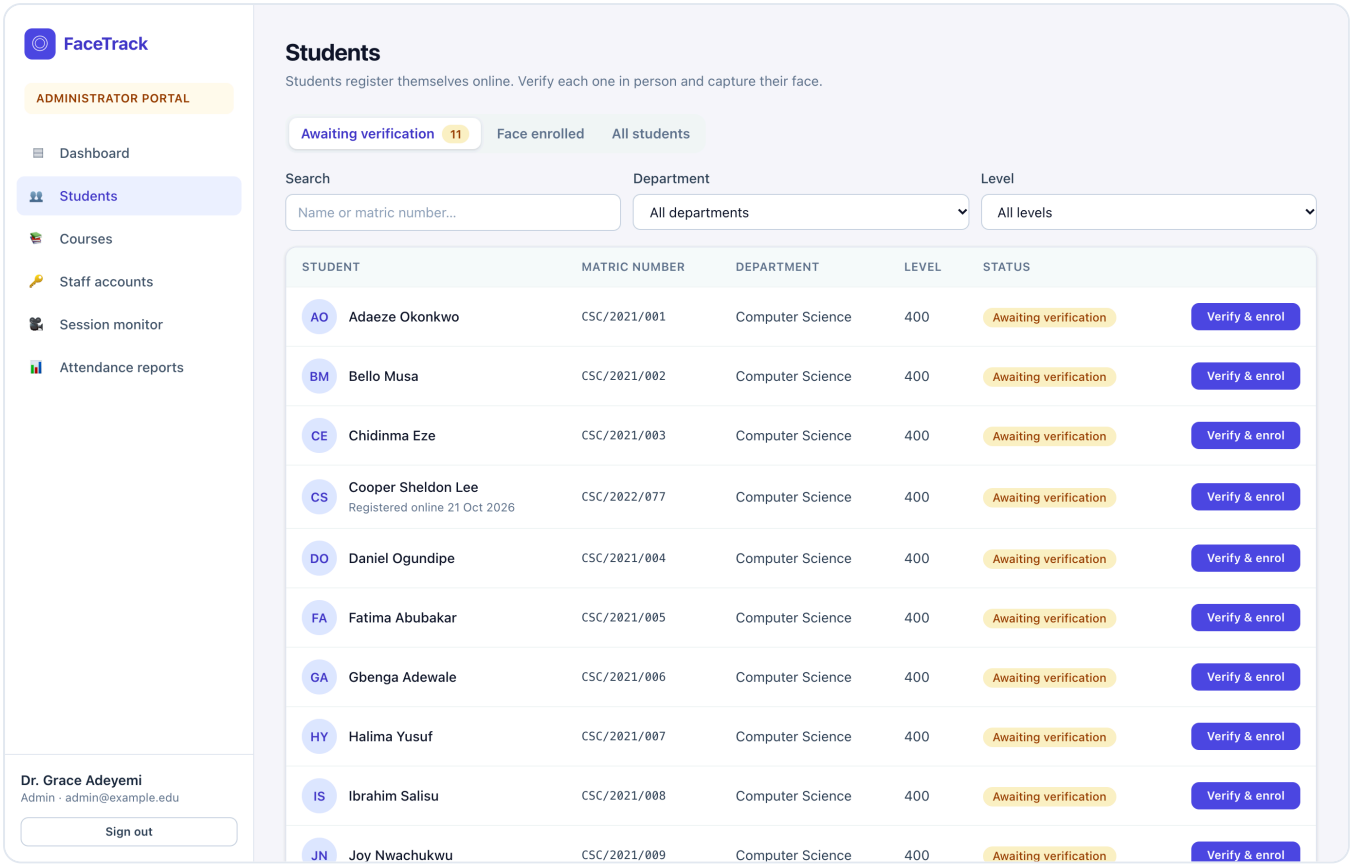


Figure 10. The verification queue. Students who registered online are labelled with the date.

5.3 Verify & enrol — the in-person step ADMIN `/students/[id]`

This is the most important screen in the system. It ties a face to a verified identity.

1. **Open the page early** — ideally while the student is walking to your desk. The camera panel shows the engine preparing (Figure 11); wait for **Start face capture**.
2. **Check the identity.** Compare the student with their uploaded passport photo (shown on the page) and the photograph on their document. Choose the document from **Document checked** (JAMB result slip, JAMB admission letter, school admission letter or student ID card); the JAMB number is pre-filled. Tick the confirmation box. The camera stays locked until you do (Figure 12).
3. **Press Start face capture** and ask the student to look at the camera. Nothing else to press: three samples are taken **automatically** as soon as a face is in view, while on-screen prompts suggest a slight turn left and right (Figure 13).
4. When the panel turns green — *Face captured in 1.4 s* — press **Verify student & save face**. Use **Retake** if you are not happy.

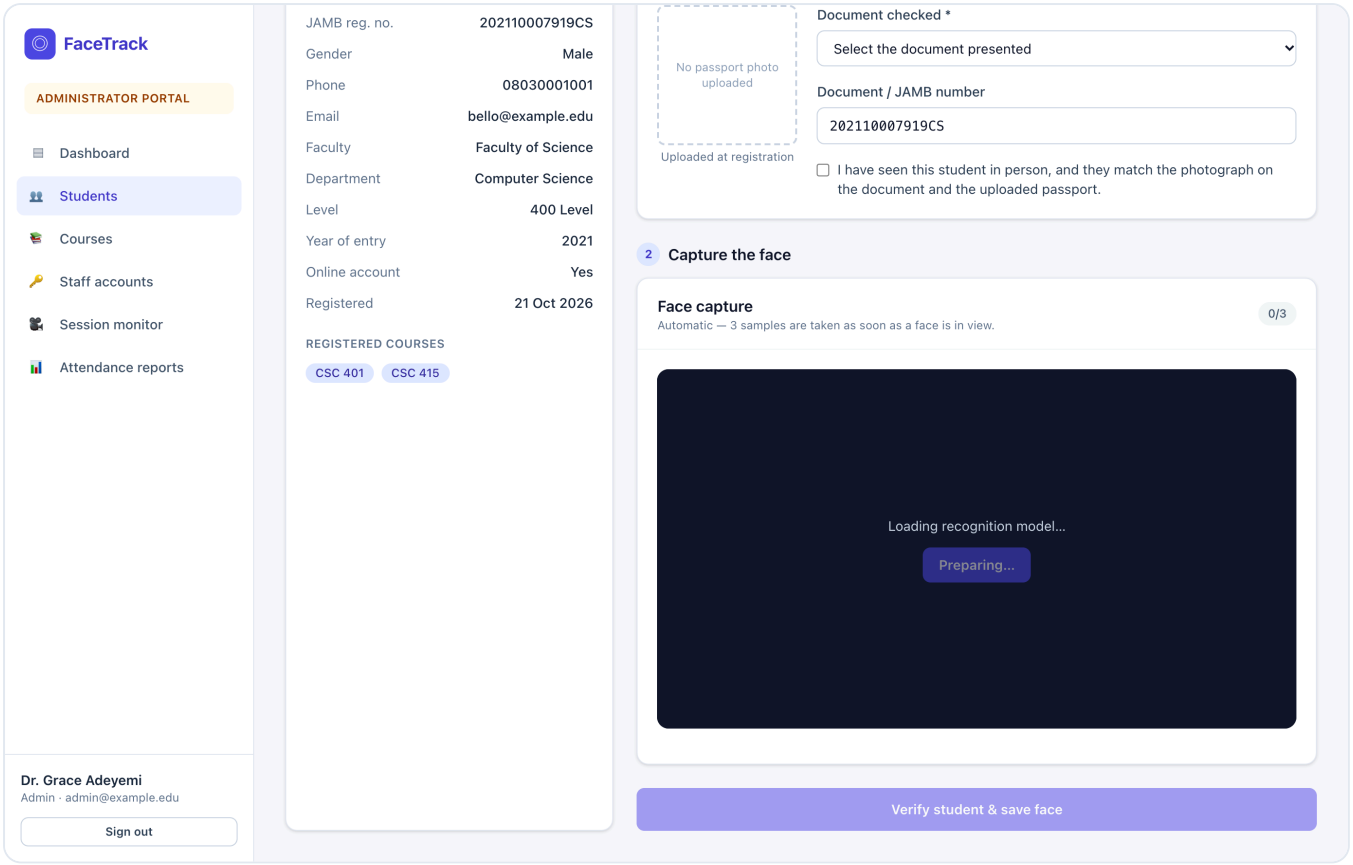


Figure 11. Just after opening the page: the engine is loading (*Loading recognition model…*, *Preparing…*). This is why the page should be opened early.

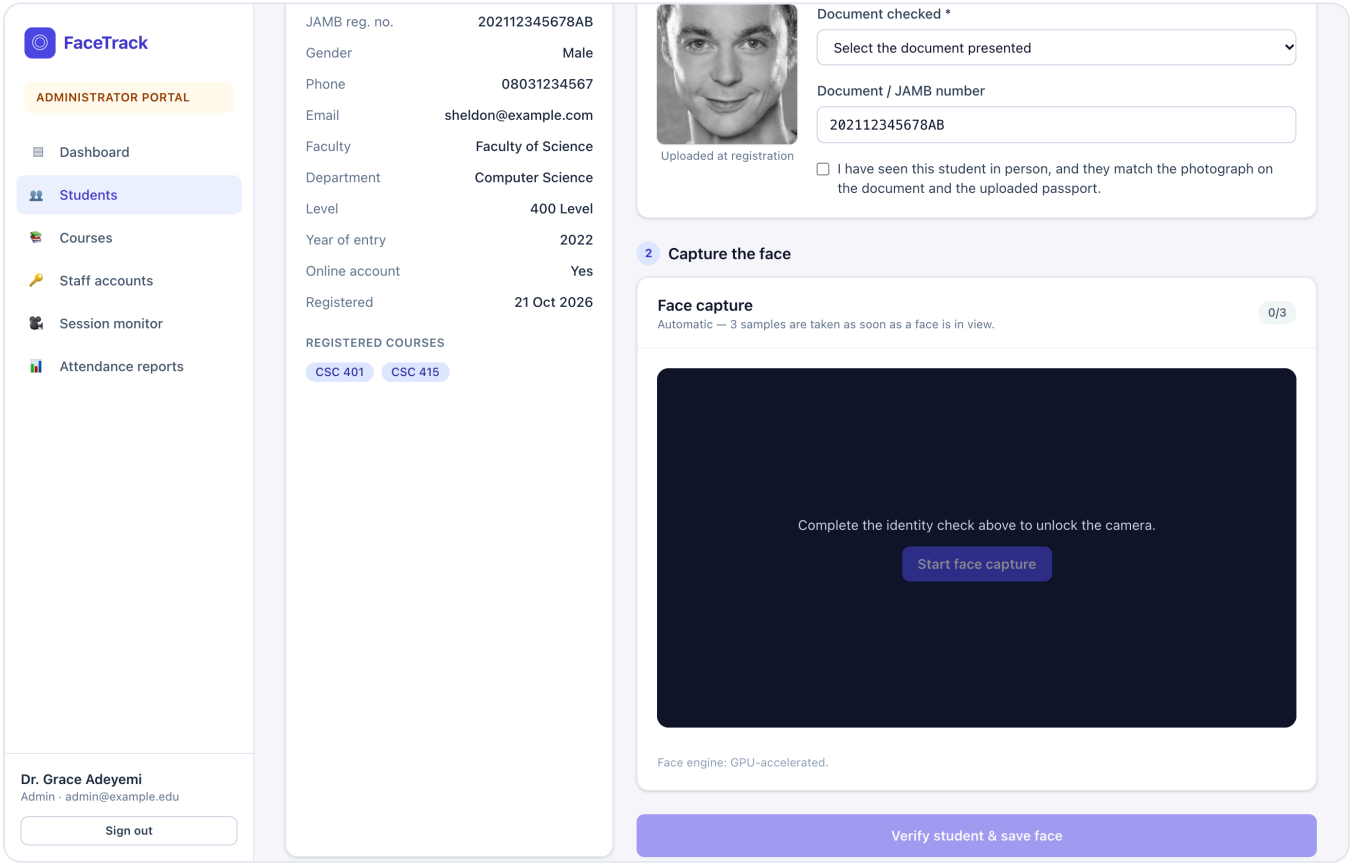


Figure 12. Engine ready, camera locked: *“Complete the identity check above to unlock the camera.”*

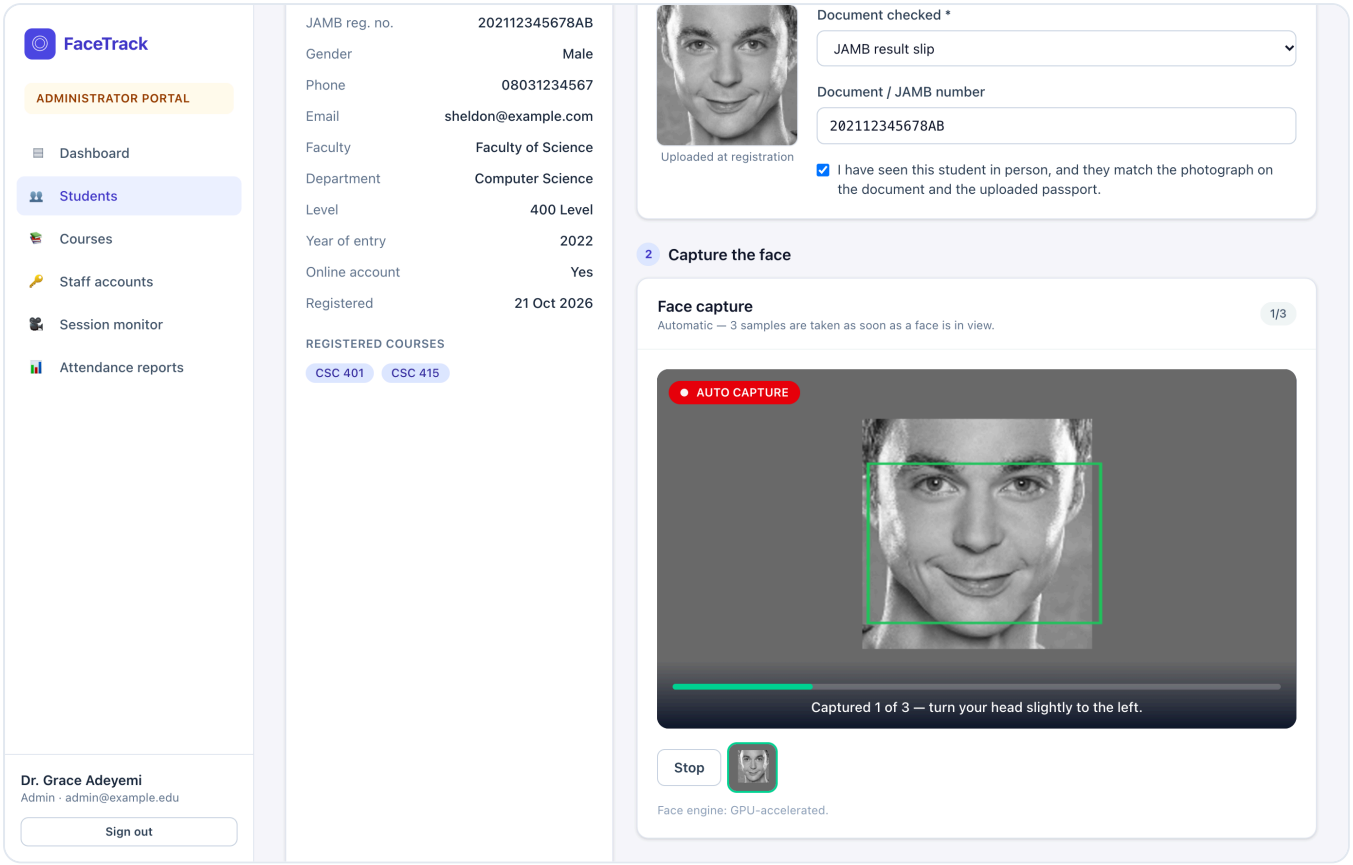


Figure 13. Automatic capture in progress — *Captured 1 of 3 — turn your head slightly to the left.* The green box shows the face being captured.

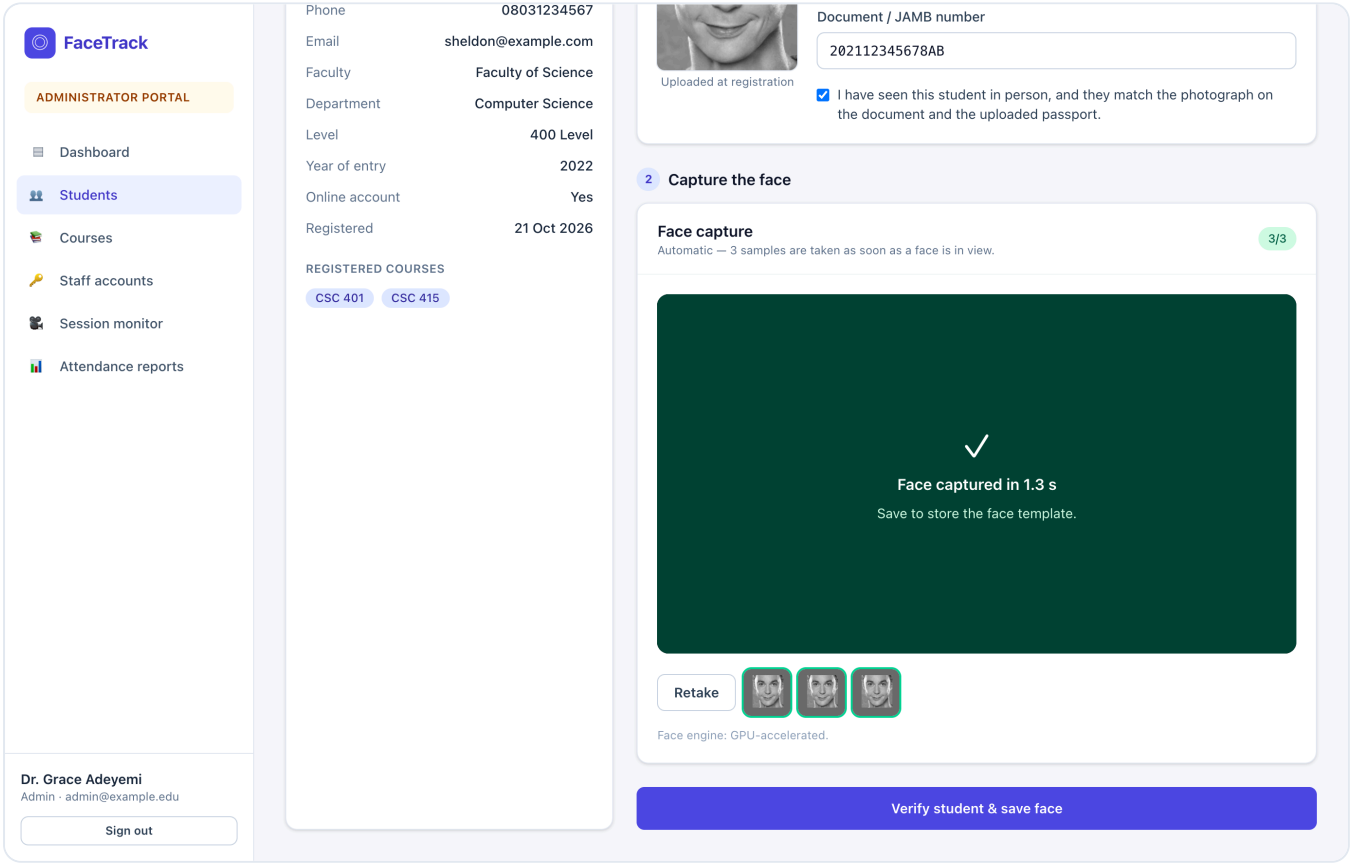


Figure 14. Done in 1.4 seconds, with the three sample thumbnails. The engine line confirms it is GPU-accelerated.

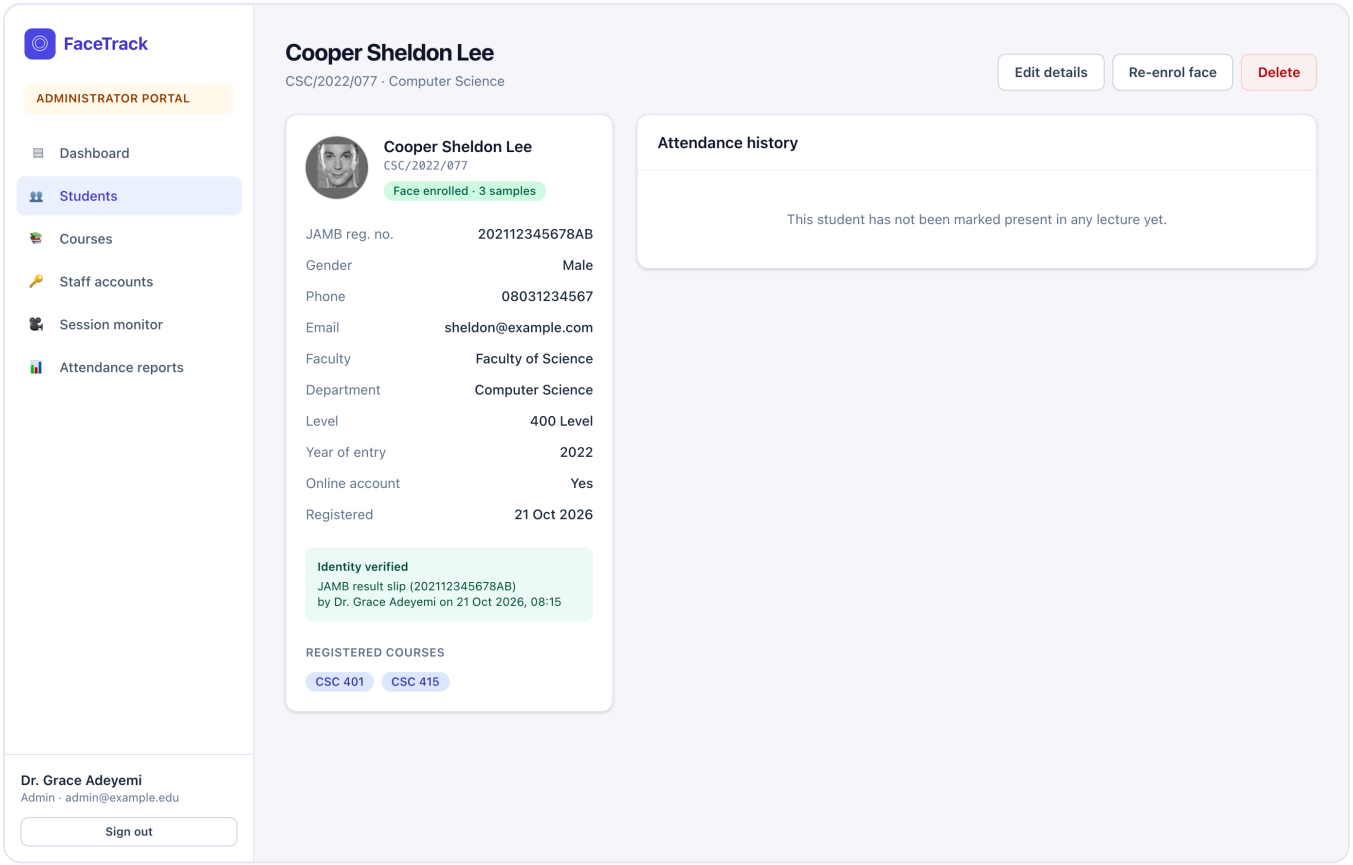


Figure 15. Saved. The profile records which document was checked, by whom and when.

> **Lighting is corrected automatically**
> Each camera frame's brightness is measured and a dim or washed-out image is corrected before detection; a note appears under the camera when this happens. Capture still works best with light on the student's face rather than behind them.

5.4 One person, one identity ADMIN

If the face being saved is already enrolled for a different student, saving is refused and the existing record is named. This blocks someone enrolling their face under a second matric number.

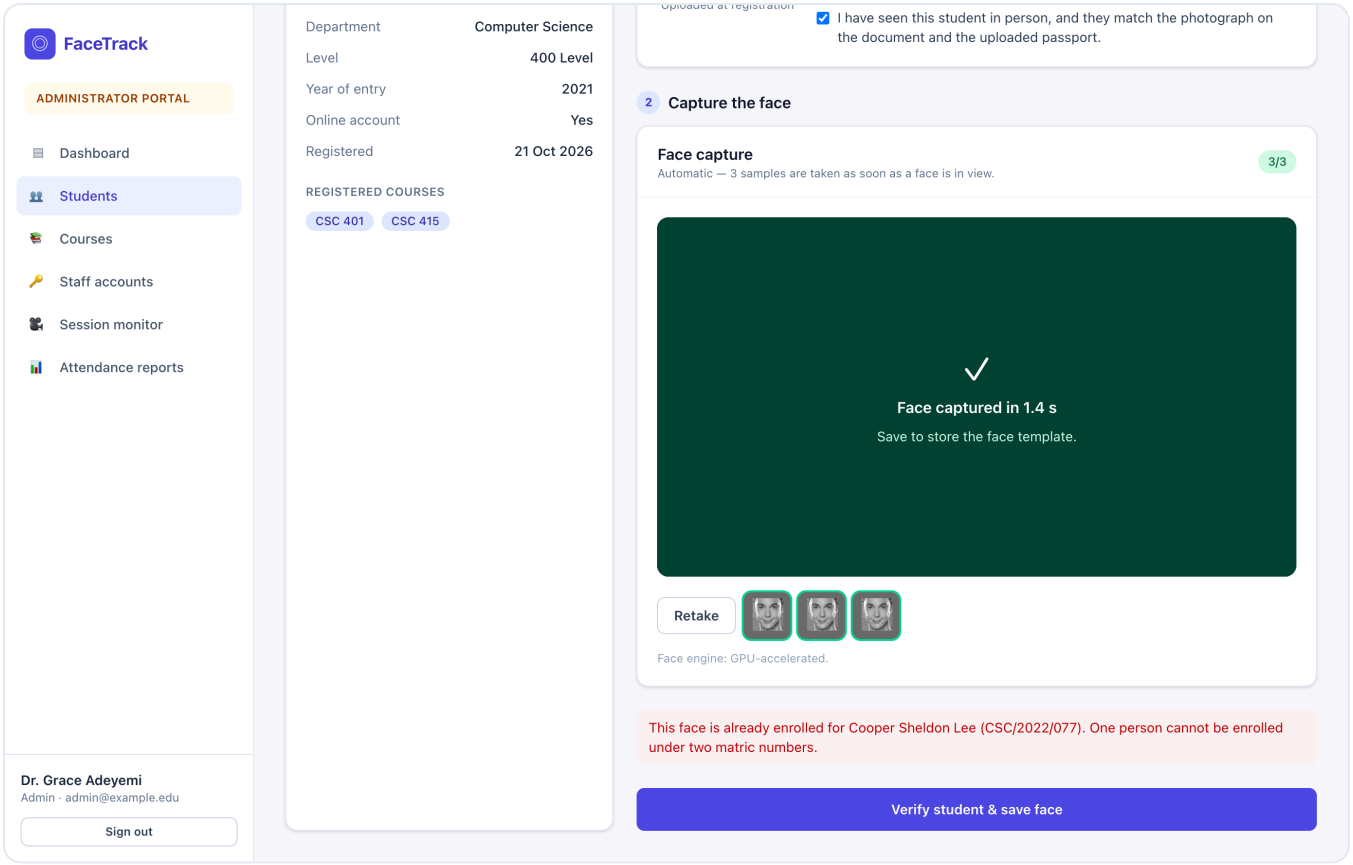


Figure 16. *"This face is already enrolled for Cooper Sheldon Lee (CSC/2022/077). One person cannot be enrolled under two matric numbers."*

5.5 Correcting details & resetting a password ADMIN

`/students/[id]`

1. Open the student and press **Edit details**.
2. Correct the matric number or name, or pick new values from the gender, faculty, department, level and year dropdowns.
3. To help a student who forgot their password, type a new one under **Reset password**.
4. Untick **Active** to stop a student being recognised (for example after withdrawal) without deleting their history. Press **Save changes**.

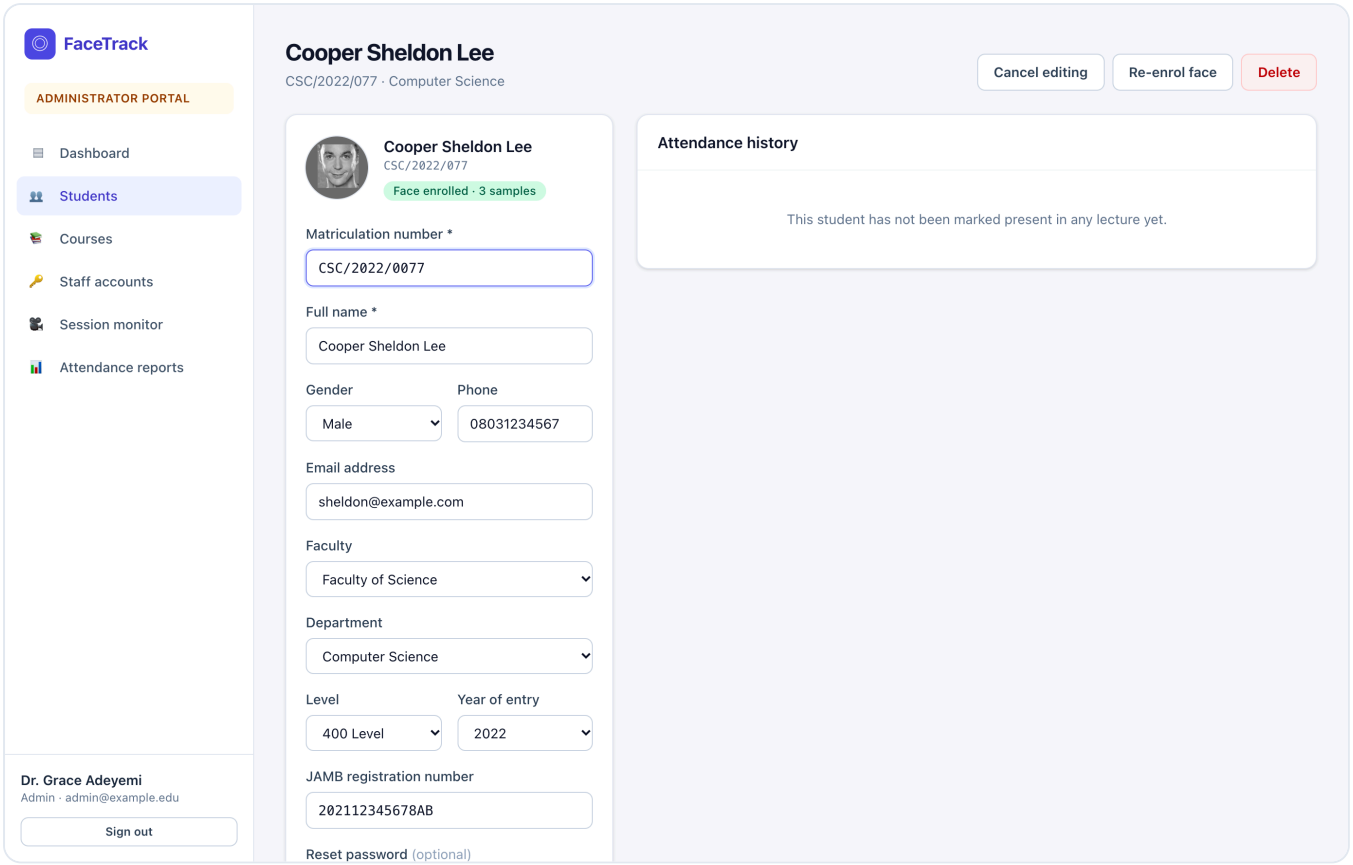


Figure 17. Editing a verified student. The face templates, courses and attendance history are kept. A matric number already used by someone else is refused.

Re-enrol face on a verified student repeats section 5.3 (identity check included) and replaces the stored templates.

5.6 Courses ADMIN

`/courses`

1. Press **+ New course**.
2. **Course code**: choose the department prefix (e.g. *CSC*) and type the three-digit number. The department fills itself in.
3. Type the **title**; pick **level**, **semester**, **credit units**, **lecturer** and **required attendance** from the dropdowns.
4. Press **Create course**. Students can now register it from their portal.

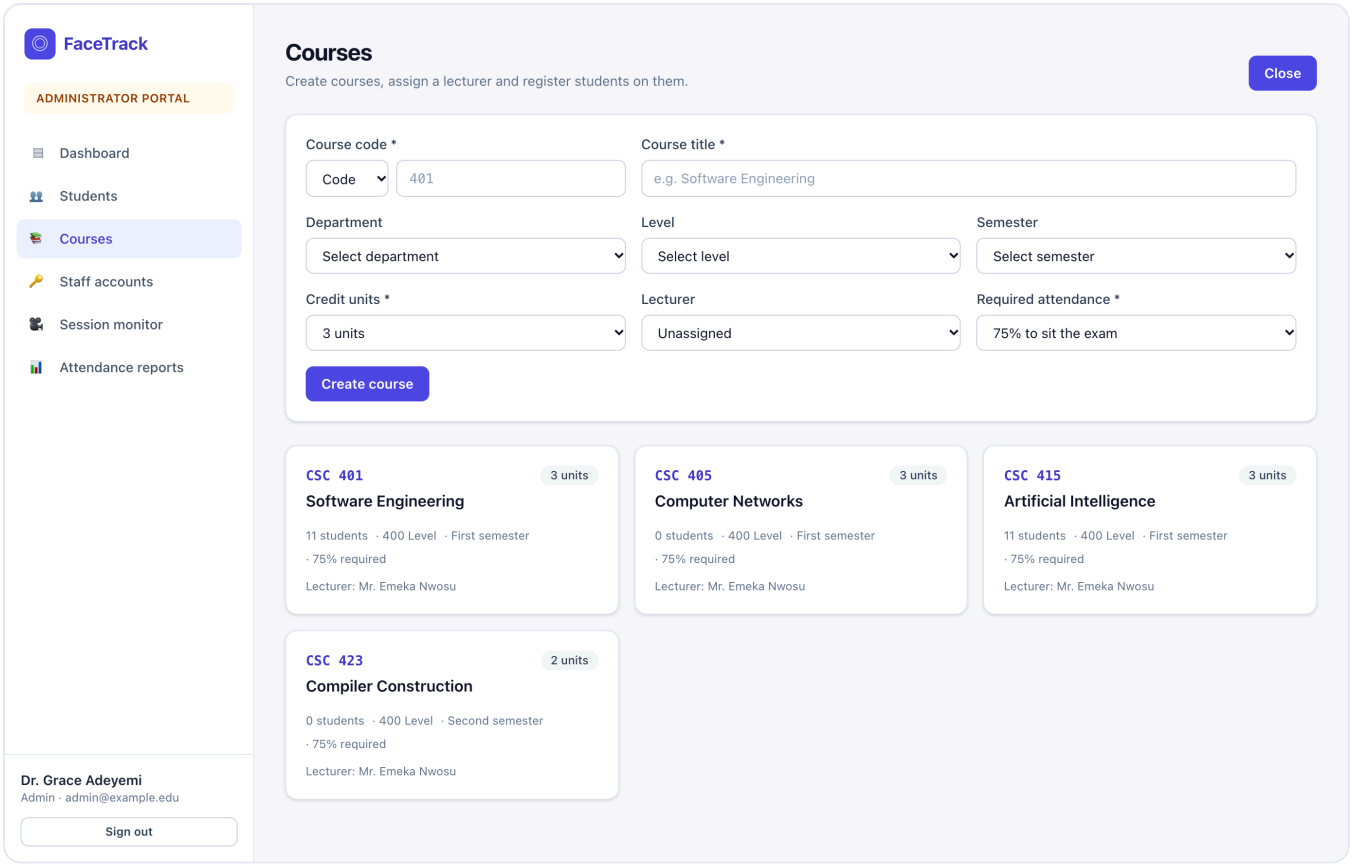


Figure 18. The new-course form. Only the course number and title are typed.

5.7 Editing a course & assigning its lecturer ADMIN `/courses/[id]`

Open a course and press **Edit course**. Pick a different **Lecturer** to hand the course over (or *Unassigned*), change any detail, and **Save changes**. Only the assigned lecturer can see the course and take its attendance.

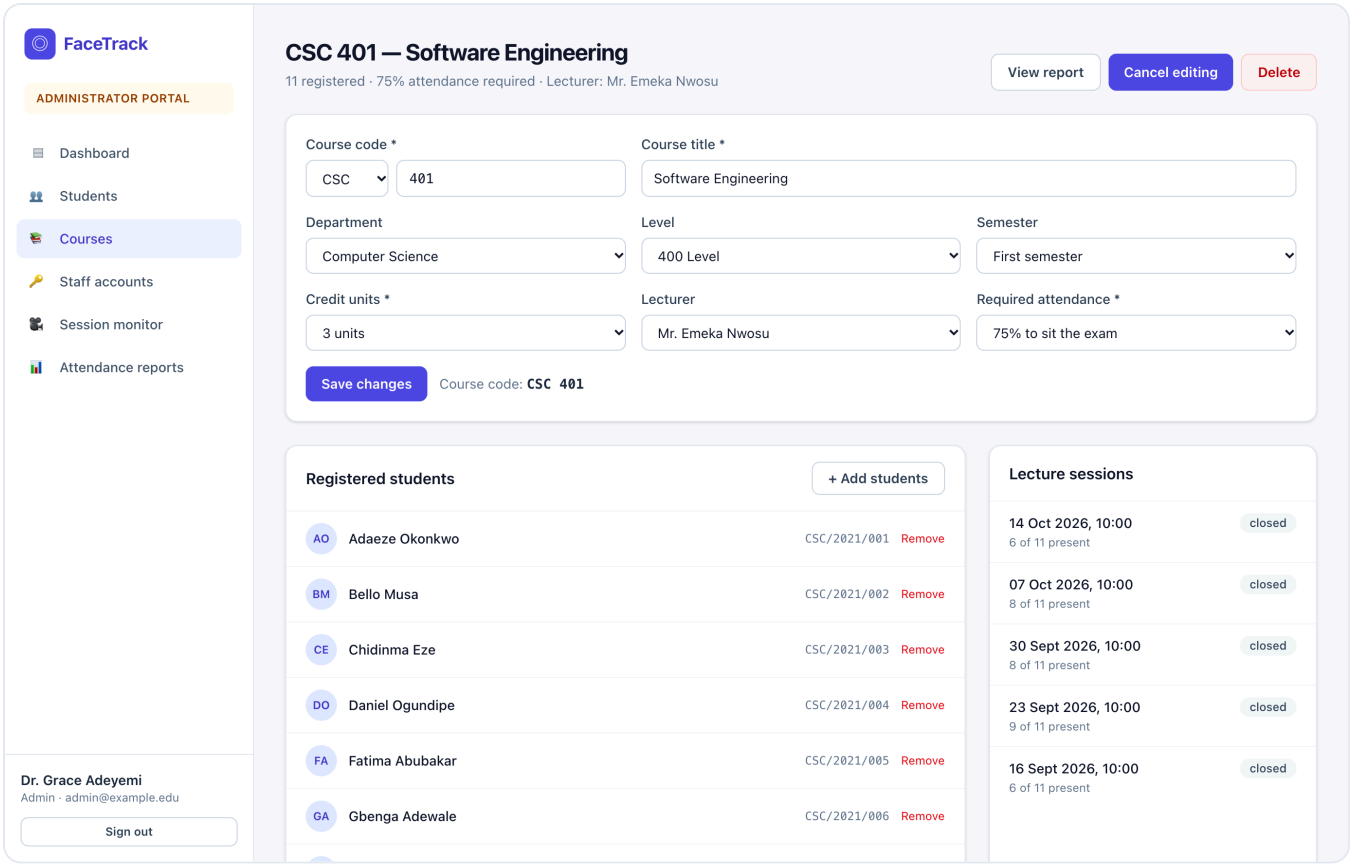


Figure 19. Editing CSC 401. **Delete** removes the course with all its sessions and attendance.

5.8 Class lists ADMIN

`/courses/[id]`

Students normally register courses themselves. The administrator can still adjust a class list: **+ Add students** opens a picker of students not yet on the course; **Remove** takes a student off. Only students on a course's class list are recognised in its lectures.

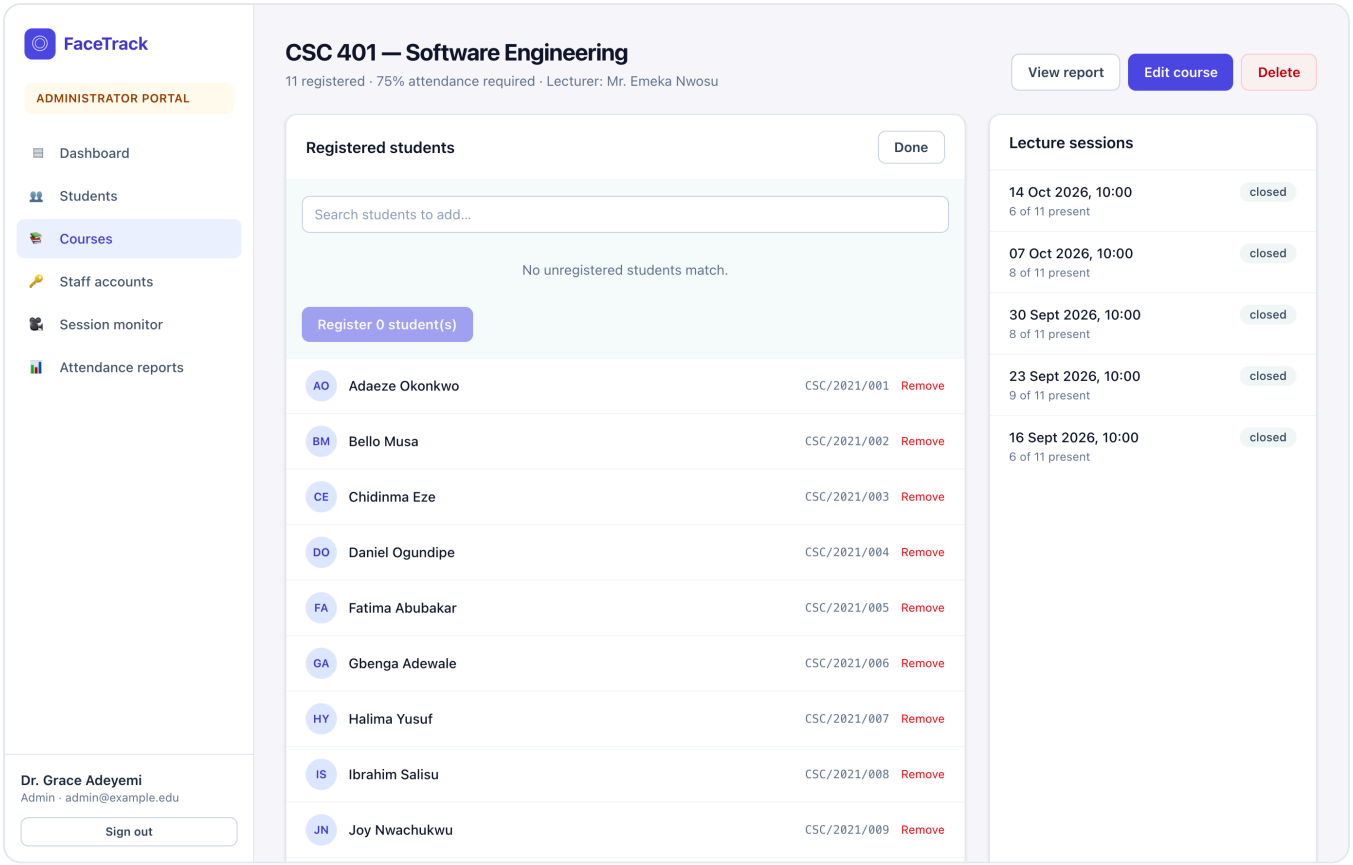


Figure 20. The add-students picker. A red *no face* badge marks students not yet verified.

5.9 Staff accounts ADMIN `/staff`

Choose a **title** (Prof., Dr., Engr., Mr., Mrs., Ms.), type the name, email and a password of at least six characters, pick the **role** and **department**, and press **Create account**. Create a lecturer here before assigning them a course.

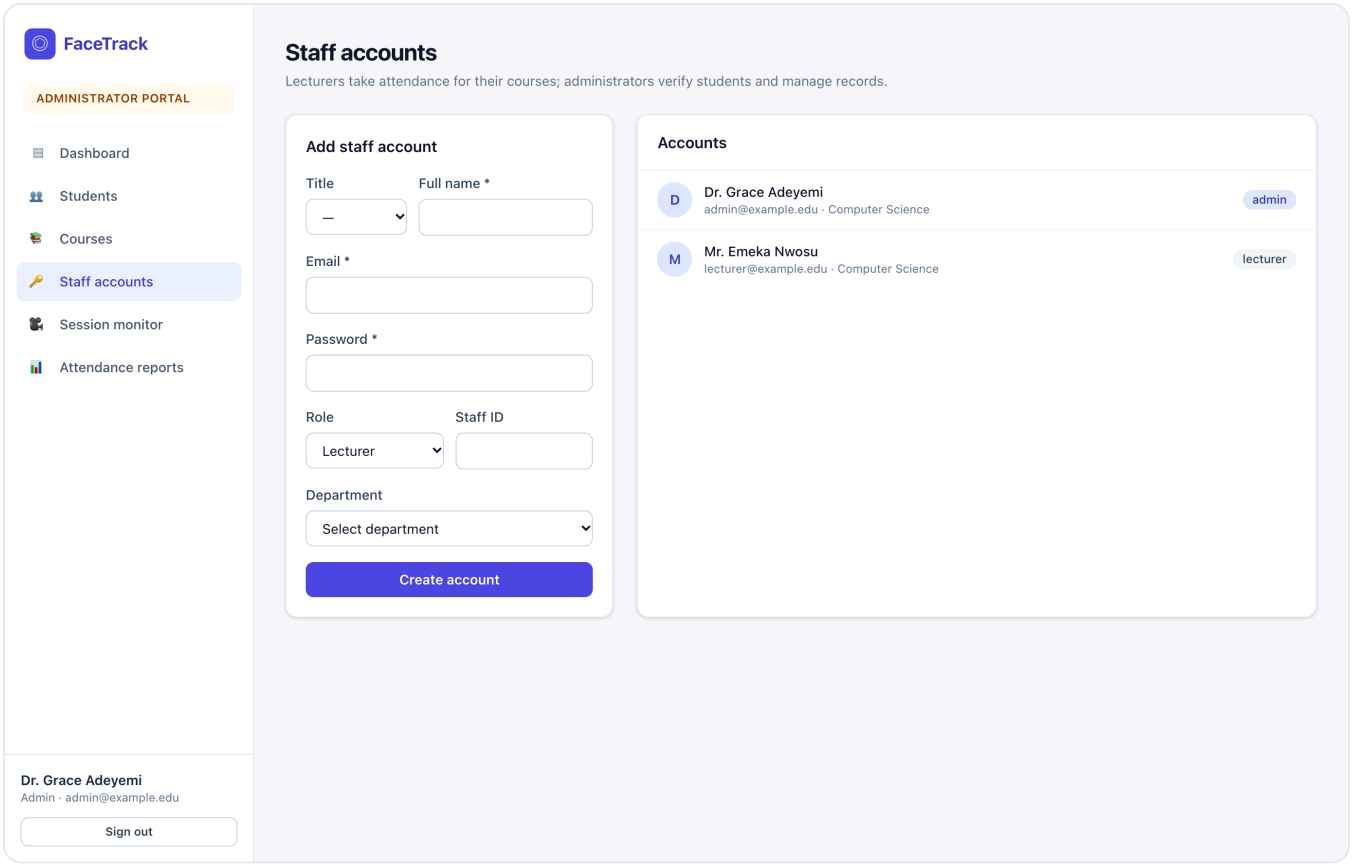


Figure 21. Staff accounts.

5.10 Session monitor & reports ADMIN

`/sessions · /reports`

The **session monitor** lists every lecture across the department — course, lecturer, time, venue and how many were present. It is read-only: **View register** opens any register, but starting, capturing and closing belong to the lecturer. **Attendance reports** shows any course as a grid (section 6.6).

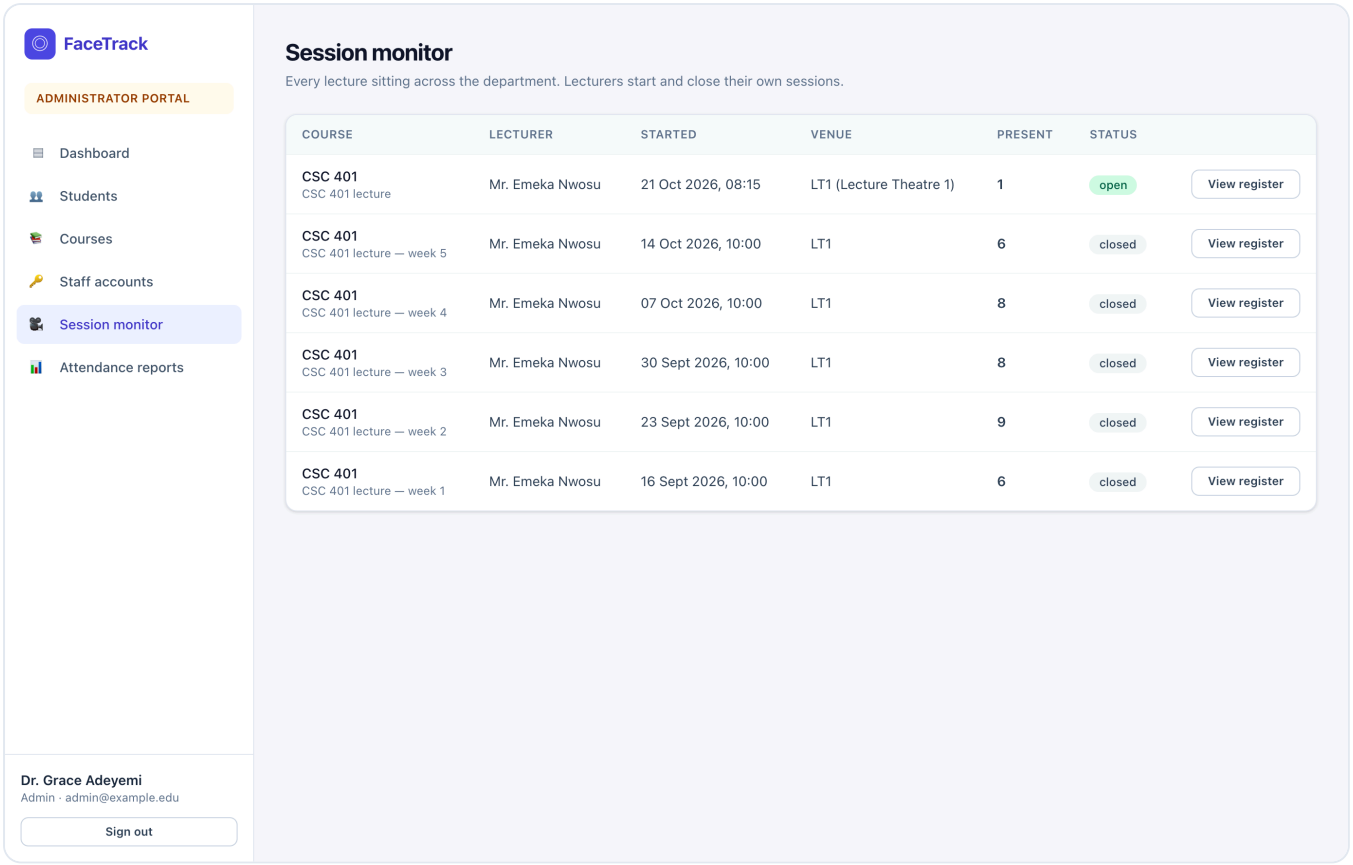


Figure 22. The session monitor, with a CSC 401 lecture in progress at the top.

6. The lecturer portal

Running a lecture and taking attendance — for the courses assigned to you.

Sidebar: Dashboard · My courses · My lecture sessions · Attendance reports

Sign in on the **Lecturer** tab.

6.1 Lecturer dashboard LECTURER `/dashboard`

Your courses, sessions held, any class in progress (with **Open camera**), attendance marked today and recent records. **Start a lecture session** is at the top right.

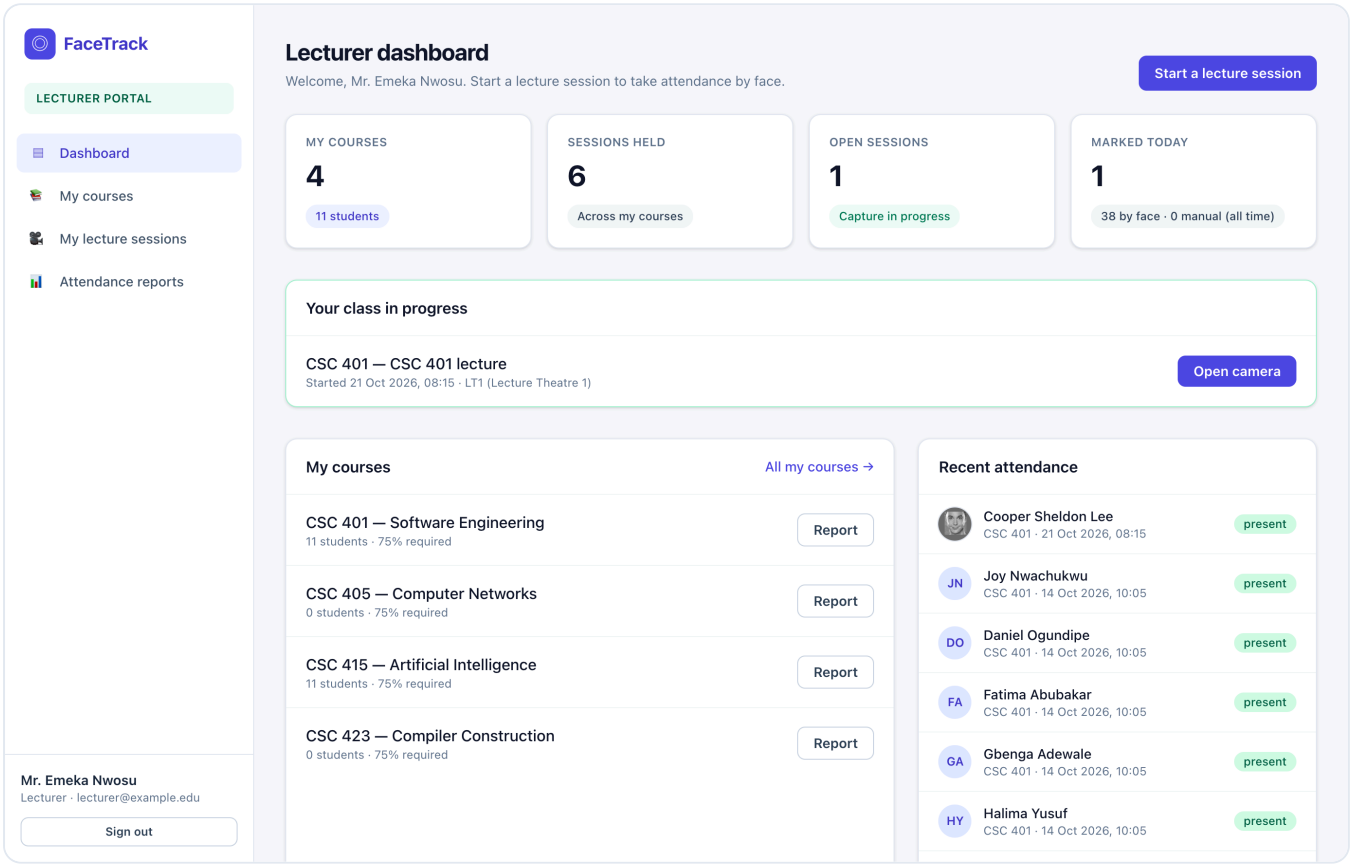


Figure 23. The lecturer dashboard with a CSC 401 class in progress.

6.2 My courses LECTURER `/courses`

The courses assigned to you. Open one to see its class list (read-only) and session history, or to **Start lecture session**. If the list is empty, ask the administrator to assign you as the course's lecturer.

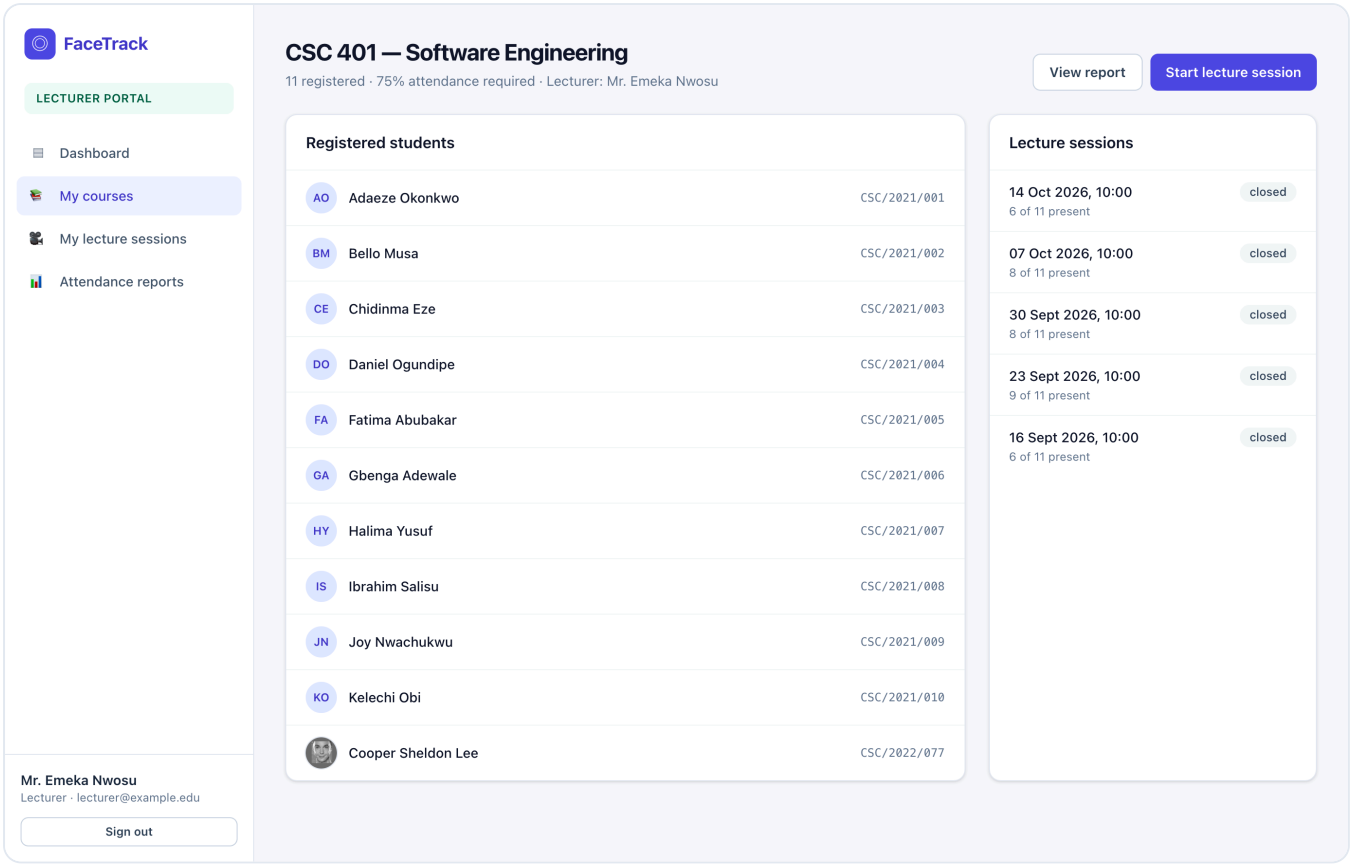


Figure 24. A course as the lecturer sees it.

6.3 Starting a lecture session LECTURER `/sessions`

1. Choose the **course**, the **venue** (LT1, LT2, laboratories…) and **Mark late after** (5–45 minutes) from the dropdowns.
2. Press **Start session & open camera**. Only one session per course can be open at a time.

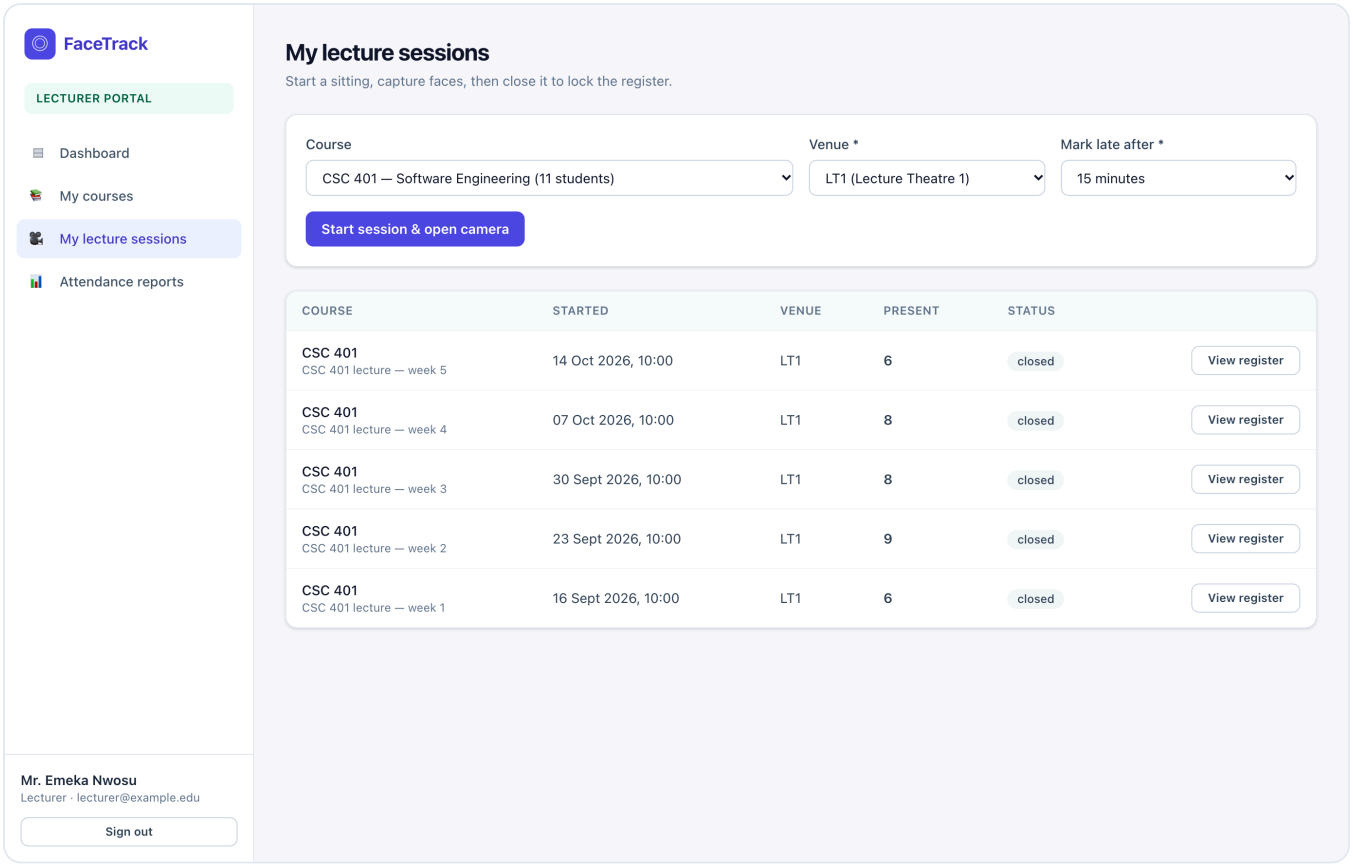


Figure 25. Every choice is a dropdown. Past sessions are listed below.

6.4 Live capture — taking attendance LECTURER `/sessions/[id]/capture`

1. Open the capture screen a minute before you need it; wait until **Start capture** is enabled.
2. Press **Start capture** and allow the camera. A red **SCANNING** badge appears; a frame is analysed about every second.
3. Point the camera at the students. Each face gets a coloured box and name label; **Marked present** fills up as they are recognised.
4. For anyone the camera cannot pick up, press **Mark manually** beside their name.
5. When the lecture ends, press **Close session** to lock the register.

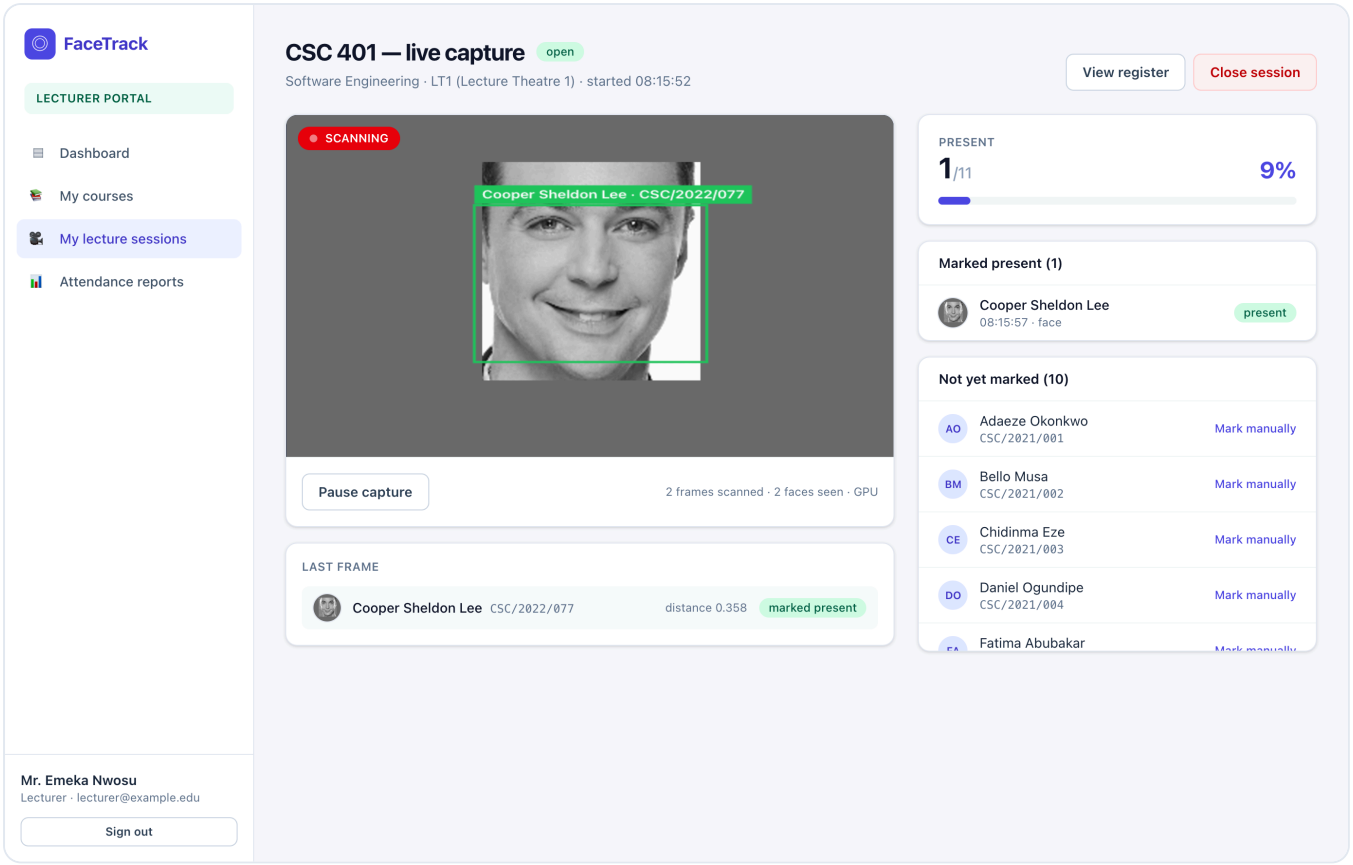


Figure 26. A verified student recognised on the first frame scanned (match distance 0.36 — below the 0.5 threshold) and marked present.

Green — newly marked
Recognised and just saved to the register.

Blue — already marked
Recognised, already recorded this session. Nothing changes.

Red — not recognised
Not verified yet, or not registered on this course.

> **The status line under the video**
> *"N frames scanned · N faces seen · GPU"* shows the system is working. If frames are not climbing, capture is paused. *CPU mode (slower)* means hardware acceleration is off (section 2.2).

6.5 Session register LECTURER

`/sessions/[id]`

Who was present, late and absent, with times and the method (face or manual — face records show the match distance). **Export CSV** downloads it; **Remove** deletes a wrong record; **Close / Reopen session** locks or unlocks it.

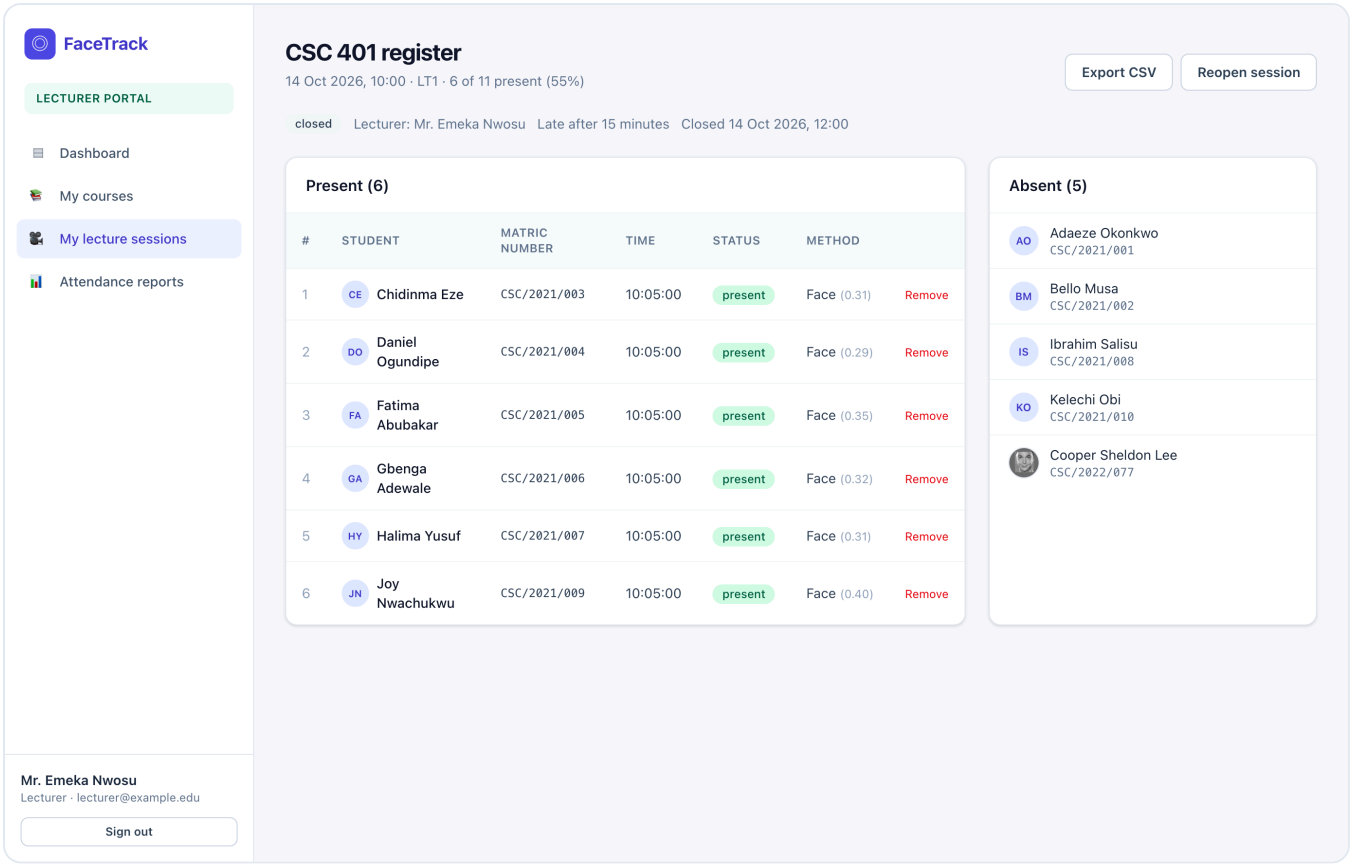


Figure 27. A closed register. The administrator sees the same page read-only.

6.6 Attendance reports LECTURER & ADMIN `/reports`

Pick a course to see every student against every lecture (**P** present, **L** late, **A** absent), each student's percentage and exam eligibility, and the summary cards. **Export CSV** for the records.

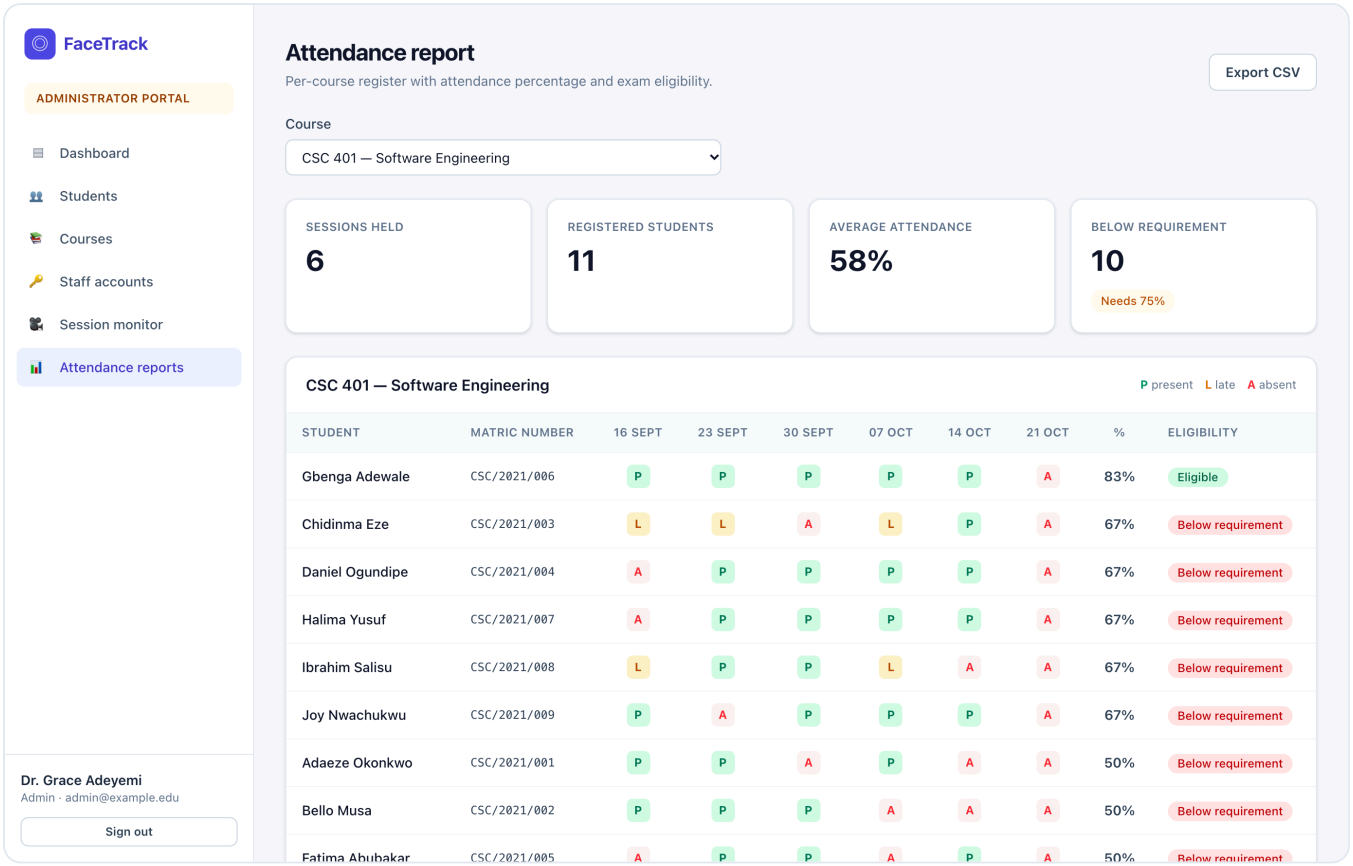


Figure 28. The CSC 401 report, best attendance first.

7. User flows

How the three portals hand work to each other.

STUDENT Flow A — Registering (from anywhere)

STEP 1 **Create account** — details, JAMB no., passport → STEP 2 **Register courses** — filtered to dept & level → STEP 3 **Visit admin office** — bring JAMB result / letter

ADMINISTRATOR Flow B — Verifying and enrolling (in person)

STEP 1 **Open page early** — engine prepares → STEP 2 **Check identity** — document + passport → STEP 3 **Auto capture** — 3 samples, ~2 s → STEP 4 **Verify & save** — duplicates refused

LECTURER Flow C — Taking attendance in a lecture

STEP 1 **Start session** — course, venue, cut-off → STEP 2 **Start capture** — allow the camera → STEP 3 **Scan the room** — green = marked → STEP 4 **Mop up** — mark manually → STEP 5 **Close session** — locks the register

ALL Flow D — Reporting

LECTURER / ADMIN **Course report** — grid, percentages, CSV ↔ STUDENT **Own dashboard** — percentage & eligibility per course

A worked example, end to end

#	PORTAL	WHERE	WHAT HAPPENS
1	Student	/register	Registers as *CSC/2022/077* with passport photo and JAMB number
2	Student	Course registration	Registers CSC 401 and CSC 415
3	Admin	Students → Verify & enrol	Opens the page early, checks the JAMB result slip, ticks the box
4	Admin	↳ Start face capture	Three samples in 1.4 s; **Verify student & save face**
5	Lecturer	My lecture sessions	CSC 401, LT1, late after 15 min → **Start session & open camera**
6	Lecturer	Live capture	**Start capture** → green box, **Marked present (1)**
7	Lecturer	↳ finish	**Close session**
8	Student	Dashboard	Sees the lecture counted, percentage and eligibility

8. Use cases

Sixteen documented scenarios. UC-02–04 belong to the student, UC-05–10 to the administrator, UC-11–15 to the lecturer.

UC-01 Sign in through a portal

All users

Steps	Choose the Student, Lecturer or Administrator tab → matric number or email → password → Sign in.
Outcome	A signed session cookie (8 hours); that portal's dashboard opens.
If it fails	A generic message for wrong details; the right details on the wrong tab are told which tab to use.

UC-02 Register online

Student

Steps	Create your account → personal details → faculty, department, level, year (dropdowns) → matric digits → JAMB number → passport photo → password.
Outcome	A student record awaiting verification; the student is signed in.
Rules	Matric and JAMB numbers must be unique; department, level and year must come from the lists; a passport photo is required. No face is captured.

UC-03 Register courses

Student

Steps	Course registration → filter by department, level, semester → Register (or Drop).
Outcome	The student joins the course's class list and can be recognised in its lectures.
Rules	A course cannot be dropped once attendance has been recorded on it.

UC-04 View own attendance & eligibility

Student

Steps	Dashboard → My attendance.
Outcome	Per course: classes held, attended, percentage, eligibility; and every individual record.

UC-05 Verify identity & enrol face

Administrator · key use case

Before	The student is present with a JAMB result slip, admission letter or ID card.
Steps	Students → Verify & enrol → compare with passport and document → choose document → tick confirmation → Start face capture → Verify student & save face.
Outcome	Three descriptors stored; the identity check (document, number, administrator, time) recorded; the student shows as verified.
Rules	Admin only. The camera is locked until the check is complete. All samples must be of one person.

UC-06 Prevent duplicate enrolment

System

Trigger	A face is saved for a student while already enrolled for another.
Outcome	Saving is refused, naming the existing student and matric number.

UC-07 Correct details / reset password

Administrator

Steps	Open the student → Edit details → correct fields or type a new password → Save changes.
Outcome	Record updated; face templates and history kept.

UC-08 Create a course & assign its lecturer

Administrator

Steps	Courses → + New course → code prefix + number, title, dropdowns → Create; later Edit course to reassign.
Outcome	The course is offered to students and appears in the assigned lecturer's portal only.

UC-09 Adjust a class list

Administrator

Steps	Open the course → + Add students / Remove.
Outcome	The recognition candidates for that course change accordingly.

UC-10 Create a staff account

Administrator

Steps	Staff accounts → title, name, email, password, role, department → Create account.
Outcome	The account can sign in through its role's tab.

UC-11 Start a lecture session

Lecturer

Steps	My lecture sessions → course, venue, late cut-off → Start session.
Rules	Only the course's own lecturer; the course must have students; one open session per course.

UC-12 Record attendance by face recognition

Lecturer · primary use case

Steps	Start capture → point the camera at students.
Outcome	Each recognised face is recorded once, stamped *present* or *late*.
Rules	Only verified, active students on the course; only within the threshold; only while the session is open.

UC-13 Mark a student manually

Lecturer

Steps	Capture screen → Not yet marked → Mark manually.
Outcome	A record tagged *manual*, distinguishable in every report.

UC-14 **Close a session** — Lecturer

Outcome	The register is frozen; further recognition is rejected.

UC-15 **Correct a wrong attendance record** — Lecturer

Steps	Session register → Remove → confirm.

UC-16 **Produce an attendance report** — Lecturer / Administrator

Steps	Attendance reports → choose course → Export CSV.
Outcome	Full register with percentages and eligibility. Lecturers: own courses; administrators: any course.

9. Troubleshooting

The problems you are most likely to meet, and what to do.

SYMPTOM	WHY IT HAPPENS	WHAT TO DO
The camera button says Preparing…	The face engine is loading and preparing (first time in this browser session).	Wait 5–30 seconds; the panel shows progress. Next time, open the page earlier.
“Face engine: CPU mode”	The browser's hardware acceleration is off.	Capture still works, more slowly. Turn on *Use graphics acceleration* in browser settings and restart the browser.
“Complete the identity check above to unlock the camera”	By design.	Choose the document checked and tick the confirmation box.
Capture says “Looking for a face”	No face in the frame, or it is very small.	Have the student face the camera at about an arm's length. Follow the on-screen prompt.
“More than one face in view”	Someone else is in the frame.	Only the student being enrolled should face the camera.
“This face is already enrolled for …”	The person is already enrolled under another matric number.	Investigate before going further — this is the impersonation check working.
“Camera permission was denied”	The browser blocked access.	Click the padlock in the address bar, set Camera to *Allow*, reload.
Wrong-tab message at sign-in	Correct details on another portal's tab.	Switch to the tab the message names.
“This record has no online account yet”	The student was created before online accounts existed.	Administrator: Edit details → Reset password.
Matric or JAMB number “already registered”	Someone registered with it before.	If it is not the student's own account, the administrator investigates and corrects.
A red box — a real student is not recognised	Not verified yet, not on this course's class list, or inactive.	Mark manually now; then verify the student or check their courses.
Lecturer sees no courses	No course is assigned to them.	Administrator: open the course → Edit course → Lecturer.
“This session is closed”	The lecture was closed.	Reopen it from the register if it was closed by mistake.

Known limits of the prototype

- **Lighting at the extremes.** Dim and washed-out frames are corrected automatically, but near-darkness or a face strongly backlit by a window can still be missed. Manual marking covers this.
- **Range.** Faces must be reasonably large in the frame; one camera at the front will not reach the back of a 300-seat hall.
- **No liveness detection.** A printed photograph held up to the camera could pass; defeating that needs blink or depth sensing, outside this prototype's scope.
- **The threshold is a trade-off.** 0.5 is a sensible default, not a certainty.

10. Demonstration checklist & quick reference

Keep this page open on the day.

Before the panel arrives

1. Use Chrome or Edge with hardware acceleration on, on a laptop with a working webcam, facing a light source.
2. Sign in as **Administrator** and open **Students** → **Verify & enrol** for the student you will enrol. Wait until the button reads **Start face capture** and the line underneath says *GPU-accelerated*. The engine is now prepared for the whole session.
3. In a second browser window (or after the enrolment), sign in as **Lecturer** and open a **live capture** screen once, so it is ready too.
4. Have a passport photo ready on the laptop for the live registration.

The demonstration, in order

1. **Student:** /register → show the dropdowns and the matric number building itself → Create my account → Course registration.
2. **Administrator:** Students → Awaiting verification → Verify & enrol → check the JAMB slip → tick → **Start face capture** (about 2 seconds) → Verify student & save face.
3. **Impersonation check:** try to enrol the same face for another student → refused.
4. **Lecturer:** My lecture sessions → Start session → **Start capture** → green box → Close session.
5. **Student:** sign back in → the lecture counts on the dashboard.

Sign-in details

PORTAL TAB	SIGN IN WITH	PASSWORD
Student	`CSC/2021/001` … `CSC/2021/010`	`student1234`
Lecturer	`lecturer@example.edu`	`lecturer1234`
Administrator	`admin@example.edu`	`admin1234`
Live link: `https://attendace-system-david.vercel.app/`		

Box colours & report symbols

newly marked · already marked · not recognised · **P** present **L** late **A** absent